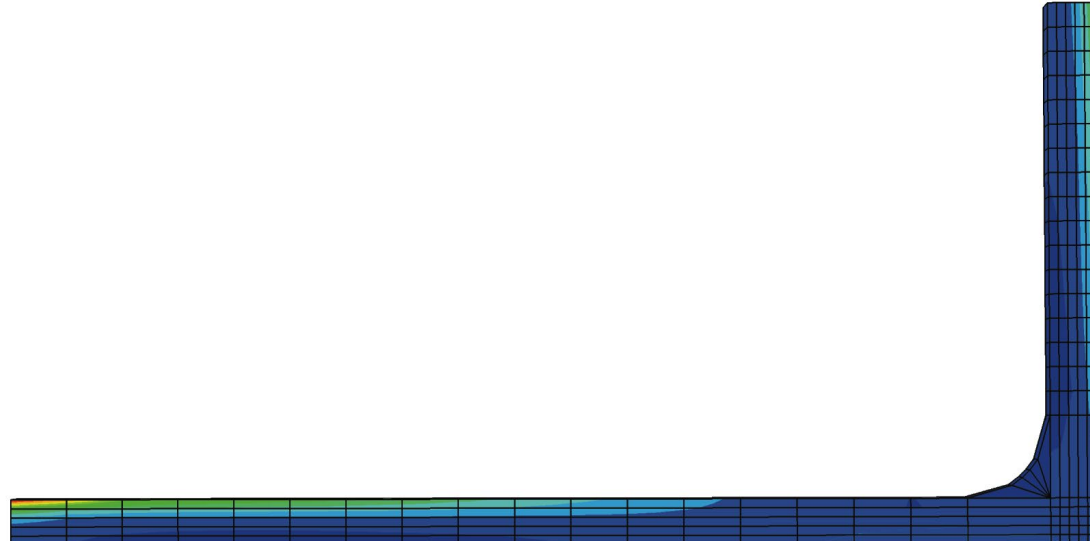


Chapter 8

Plate Elements

Book Chapters

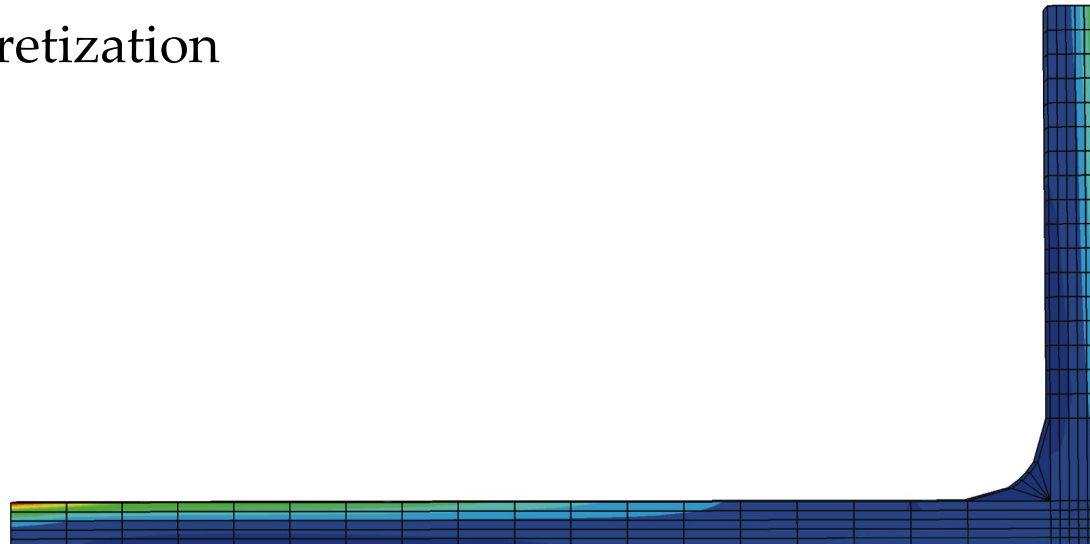
[O] V2/Ch5 & Ch6



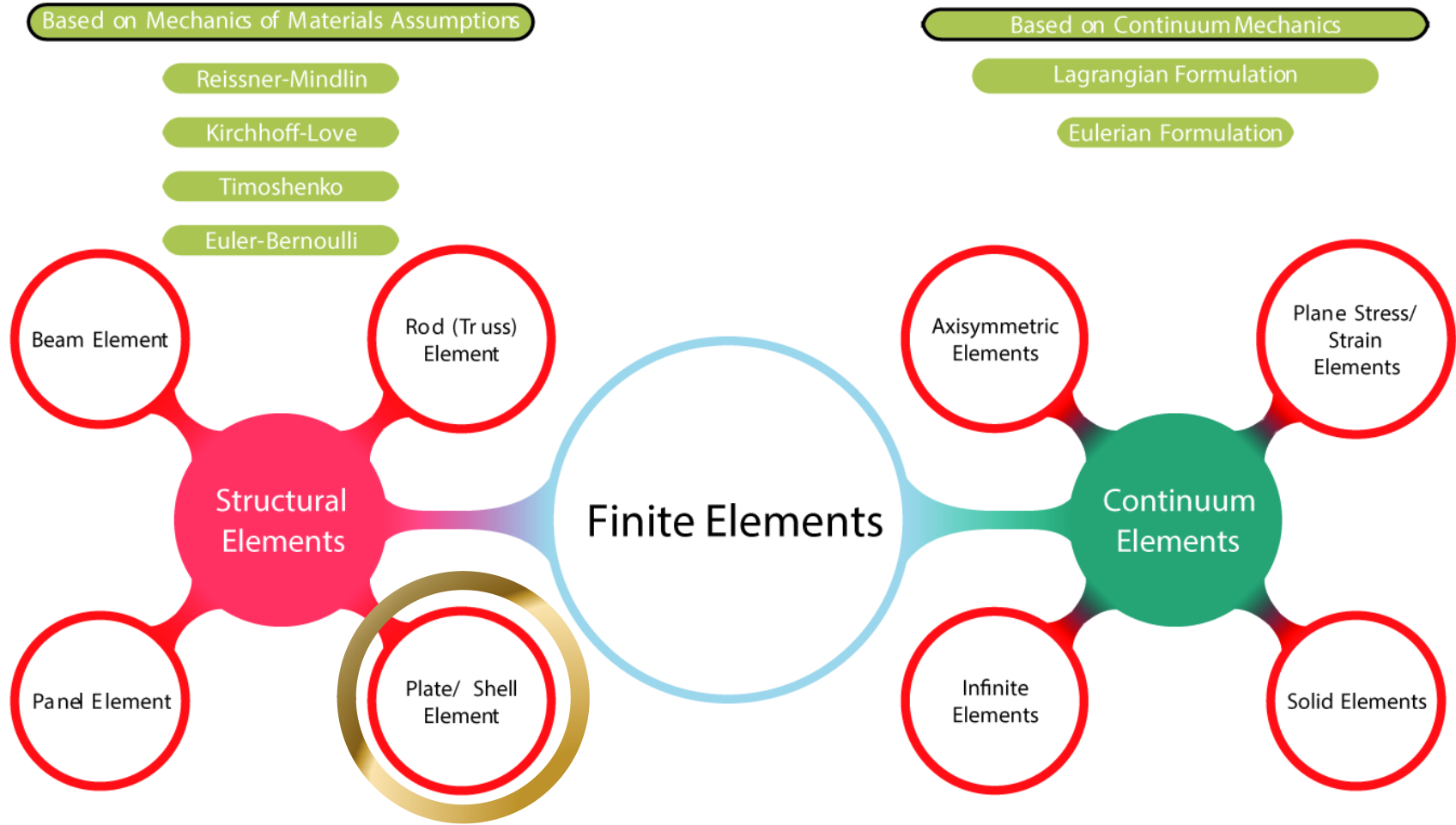
Today's Lecture

Extending (discretization) to higher Dimensions

- Plate Elements
 - Review of 3D elasticity, plane stress
 - Plate/beam theories
 - Weak form
 - Finite Element Discretization
 - Shear locking

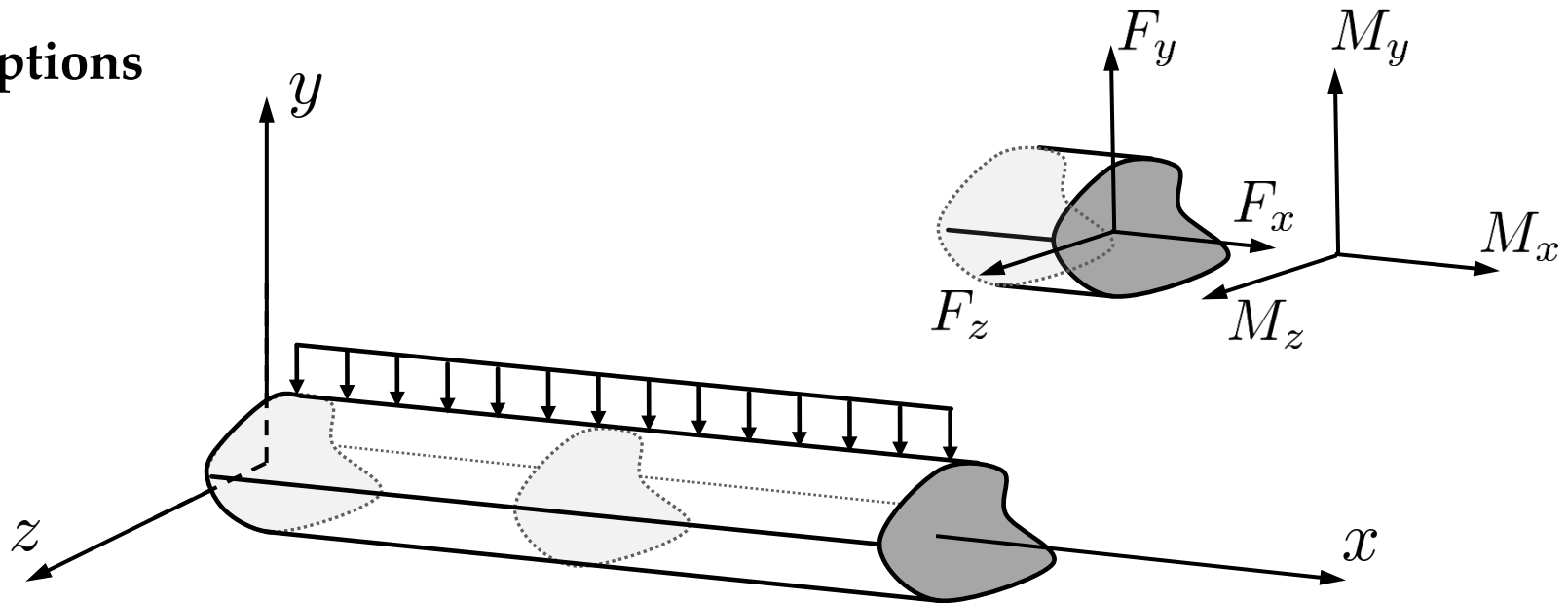


Taxonomy of Finite Elements



The Euler/Bernoulli beam theory

Assumptions



❖ Uniaxial Element

- The longitudinal direction is sufficiently larger than the other two

❖ Prismatic Element

- The cross-section of the element does not change along the element's length

❖ Euler/ Bernoulli assumption

- Upon deformation, plane sections remain plane **AND** perpendicular to the beam axis

But what happens if two dimensions are sufficiently larger than the third one?

3D elasticity problem

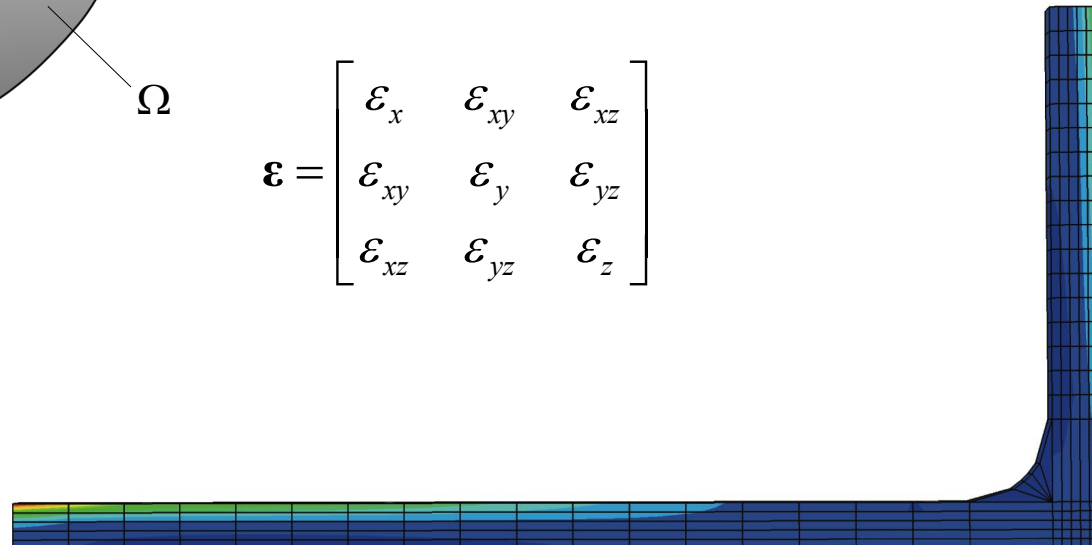
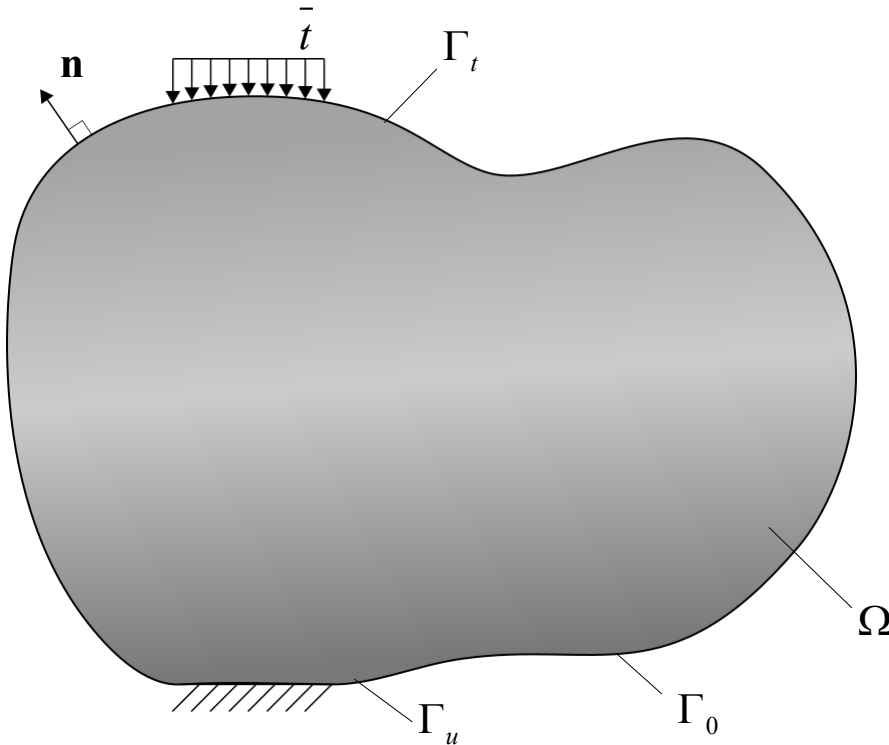
Problem variables:

Displacements:

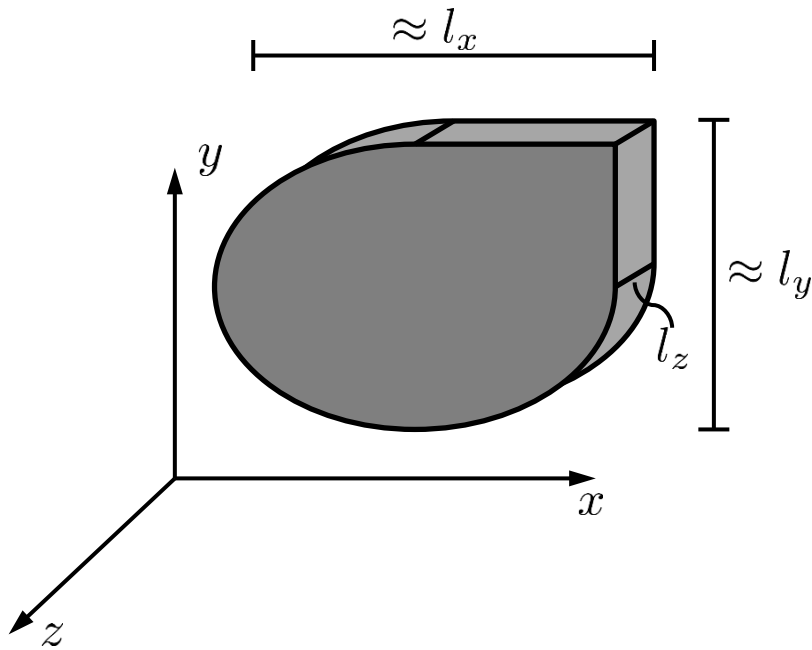
$$\mathbf{u} = \begin{Bmatrix} u_x \\ u_y \\ u_z \end{Bmatrix}$$

Strain tensor:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{xy} & \varepsilon_y & \varepsilon_{yz} \\ \varepsilon_{xz} & \varepsilon_{yz} & \varepsilon_z \end{bmatrix}$$



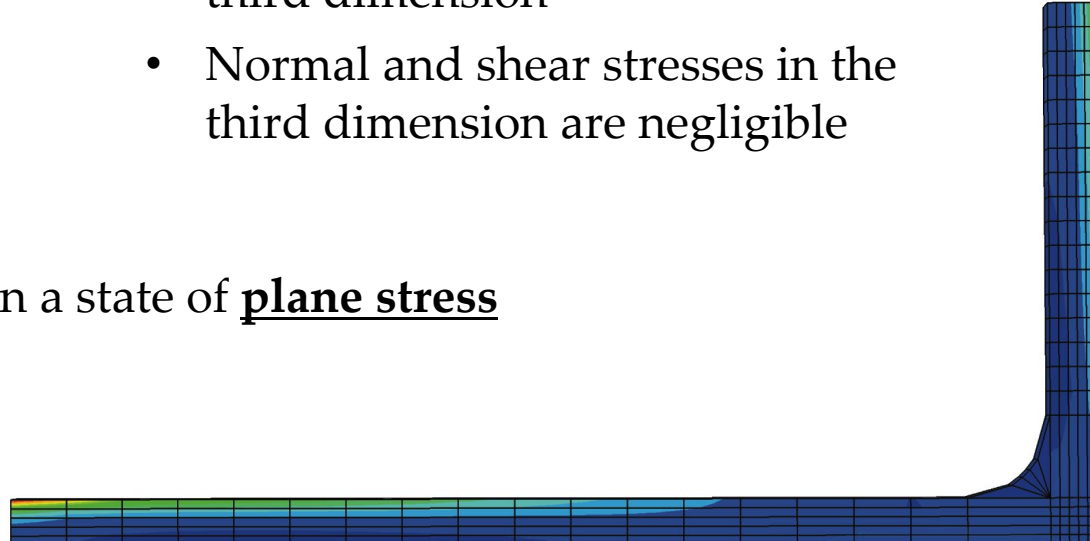
Plane stress assumptions



Consider a structure where:

- The length in one dimension is much smaller than the other two $l_z \ll (l_x, l_y)$
- Loads are applied only within a plane
- The in-plane stresses, strains and displacements are constant in the third dimension
- Normal and shear stresses in the third dimension are negligible

➤ Such a structure is said to be in a state of **plane stress**



Plane stress equations

Assuming that stresses along the third dimension are zero:

Assumptions:

$$\sigma_x = \sigma_x(x, y)$$

$$\sigma_y = \sigma_y(x, y)$$

$$\tau_{xy} = \tau_{xy}(x, y)$$

$$\sigma_z = \tau_{xz} = \tau_{yz} = 0$$

Equilibrium equations:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + F_x = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + F_y = 0$$

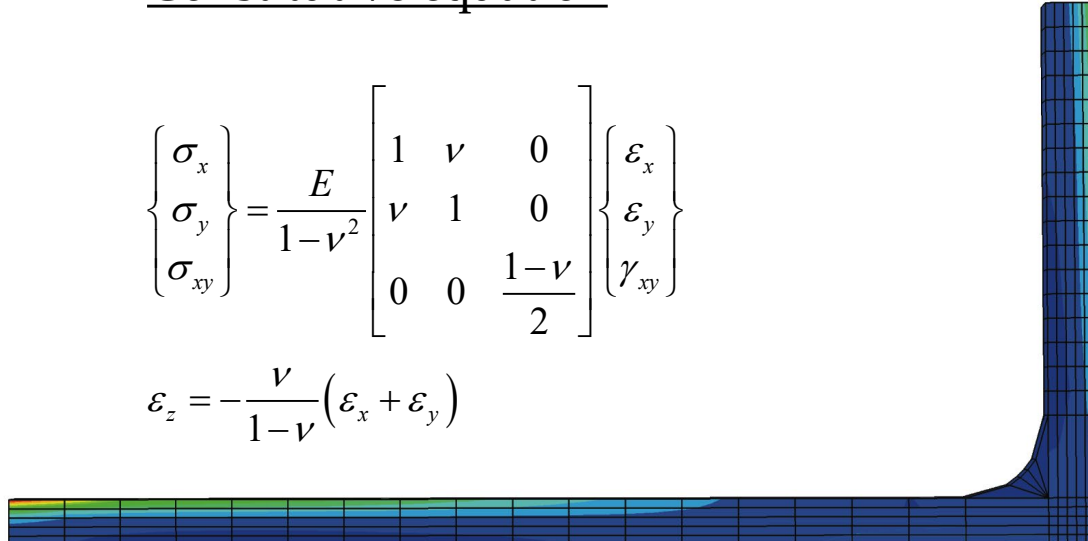
Strain definition

$$\boldsymbol{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u_x}{\partial x} \\ \frac{\partial u_y}{\partial y} \\ \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \end{Bmatrix}, \quad \begin{aligned} \varepsilon_z &= \frac{\partial u_z}{\partial z} \\ \varepsilon_{xz} &= 0 \\ \varepsilon_{yz} &= 0 \end{aligned}$$

Constitutive equation

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

$$\varepsilon_z = -\frac{\nu}{1-\nu} (\varepsilon_x + \varepsilon_y)$$

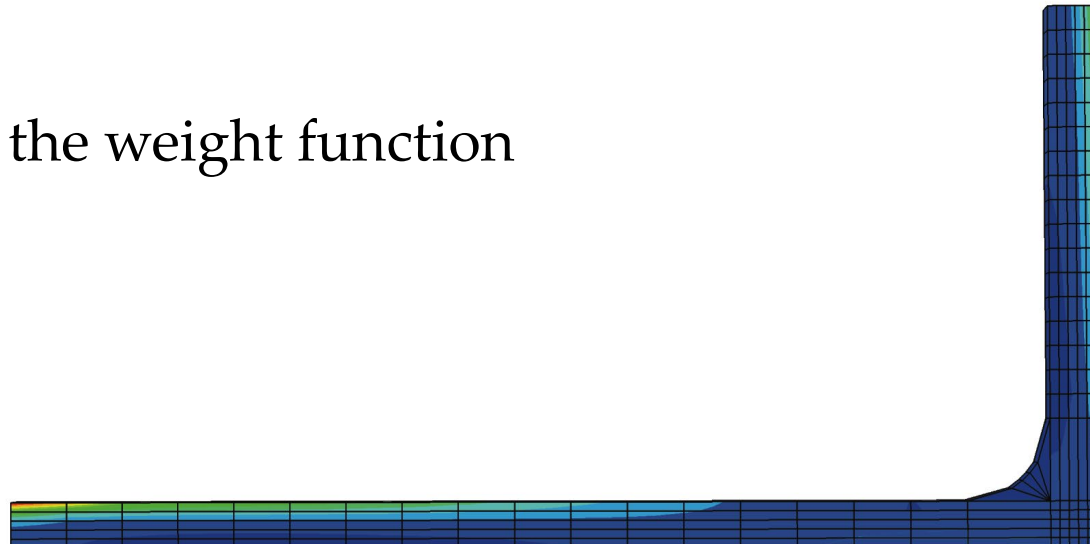


Weak form

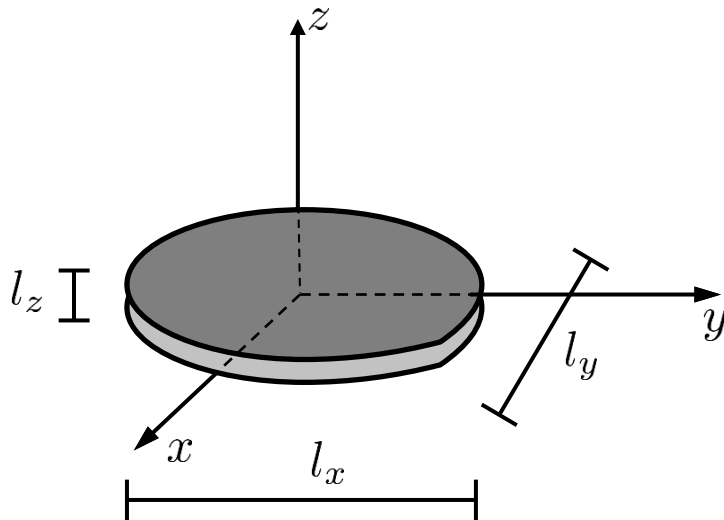
The weak form of the problem can be obtained using the Galerkin method as:

$$\int_{\Omega} \boldsymbol{\varepsilon}^T(\mathbf{w}) \mathbf{E} \boldsymbol{\varepsilon}(\mathbf{u}) d\Omega = \int_{\Omega} \mathbf{w}^T \cdot \mathbf{F} d\Omega + \int_{\Gamma} \mathbf{w}^T \cdot \bar{\mathbf{t}} d\Gamma$$

where $\mathbf{w} = \begin{Bmatrix} w_x \\ w_y \\ w_z \end{Bmatrix}$ is the weight function



The plate problem

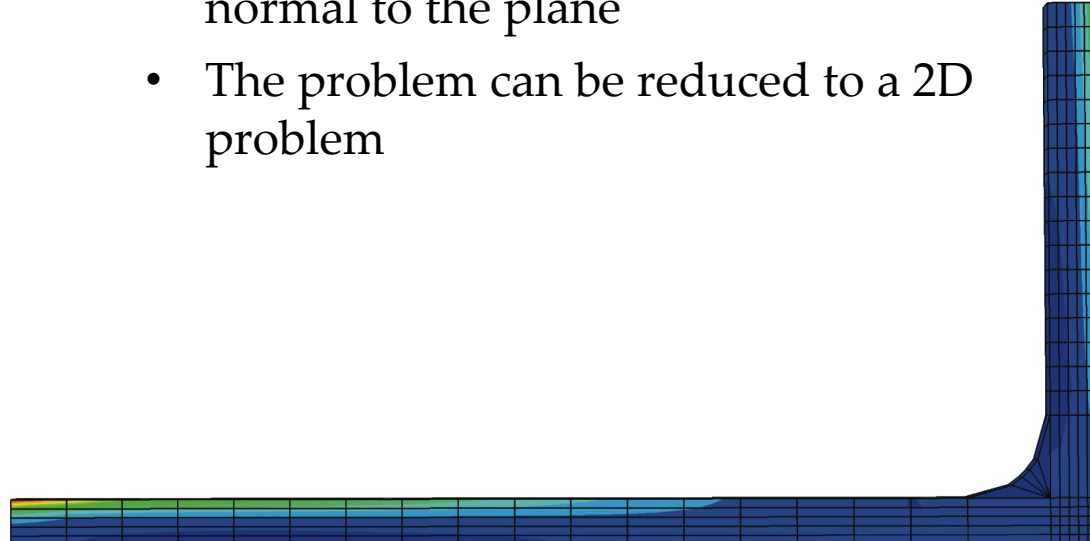


We consider structures where:

- One dimension (thickness) is sufficiently smaller than the other two dimensions

$$\frac{l_z}{\max(l_x, l_y)} < 0.1$$

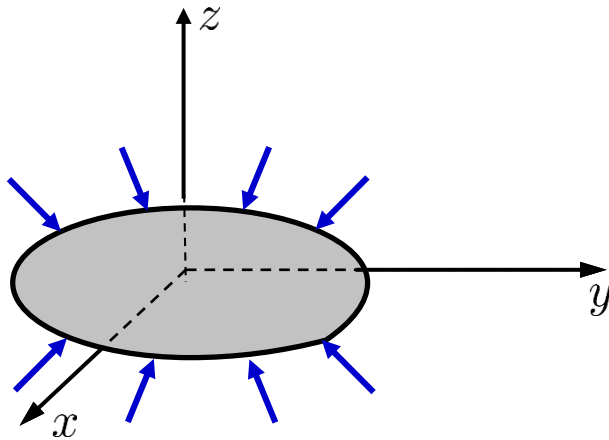
- The geometry is planar
- Loads are applied both in plane and normal to the plane
- The problem can be reduced to a 2D problem



The plate problem

Depending on the loading, the problem can be reduced to two separate problems:

In plane loading



Plane stress (or
membrane) problem

Normal loading

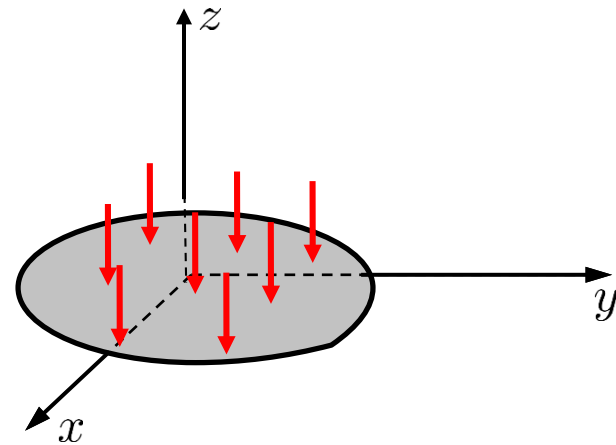
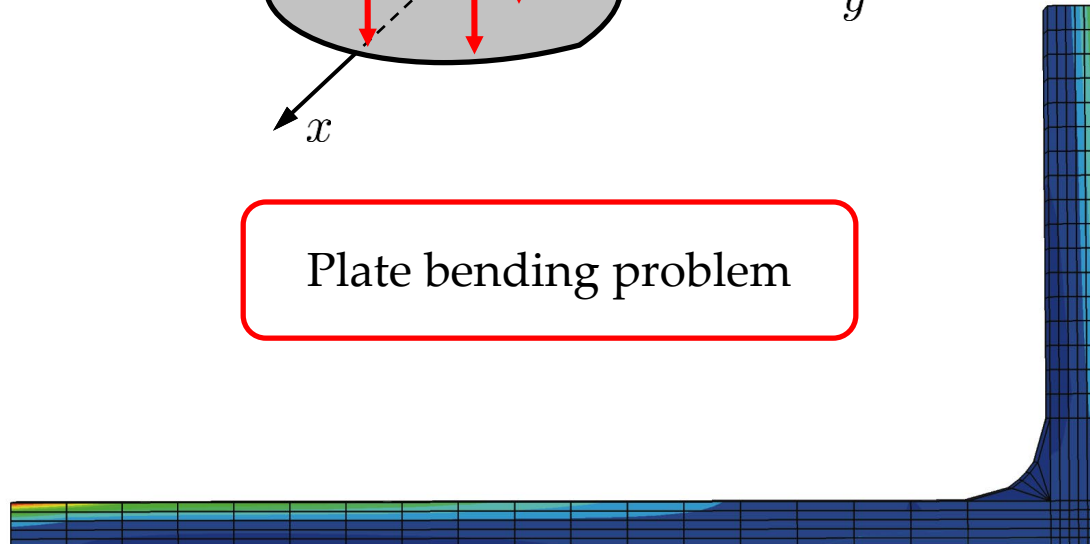
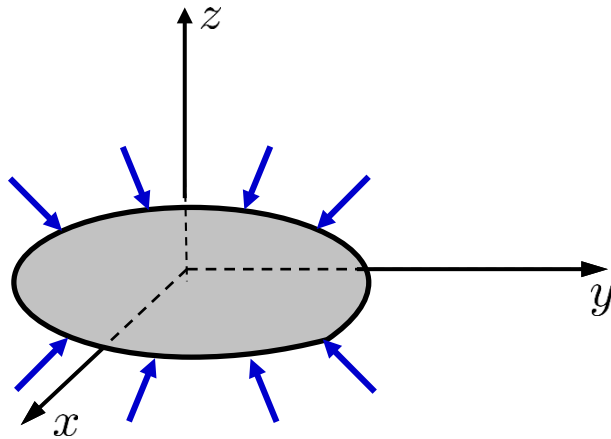


Plate bending problem



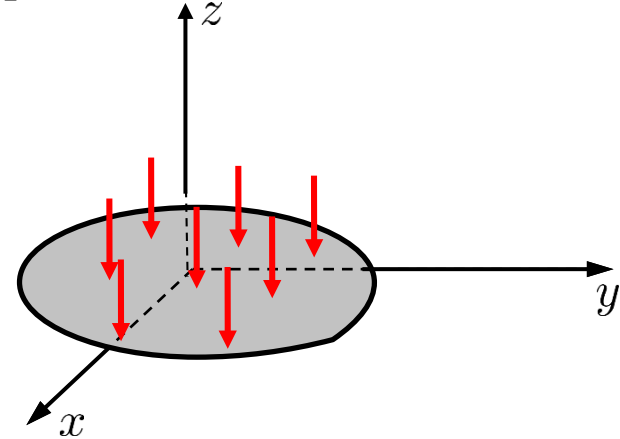
The plate problem

Case 1: Loads applied within the xy plane



Plane stress (or
membrane) problem

Case 2: Loads applied
perpendicular to the mid-surface



Bending

- i. Plate theory is only concerned with the response of the body due to bending loads
- ii. The combined response under membrane and bending conditions of plane (or curved) surfaces is treated under the framework of *shell* theory

Plate theories

Two alternative theories exist for the bending of plates which are in complete analogy to beam theories:

Beam theory

Bernoulli-Euler

Kinematic assumptions

Plane sections that are normal to the axis of the beam remain plane and normal to the axis.

Valid for

Beams with small height to length ratio

Plate theory

Kirchhoff-Love

Kinematic assumptions

Straight lines that are normal to the mid-surface remain straight and normal to the mid-surface.

Valid for

Thin plates

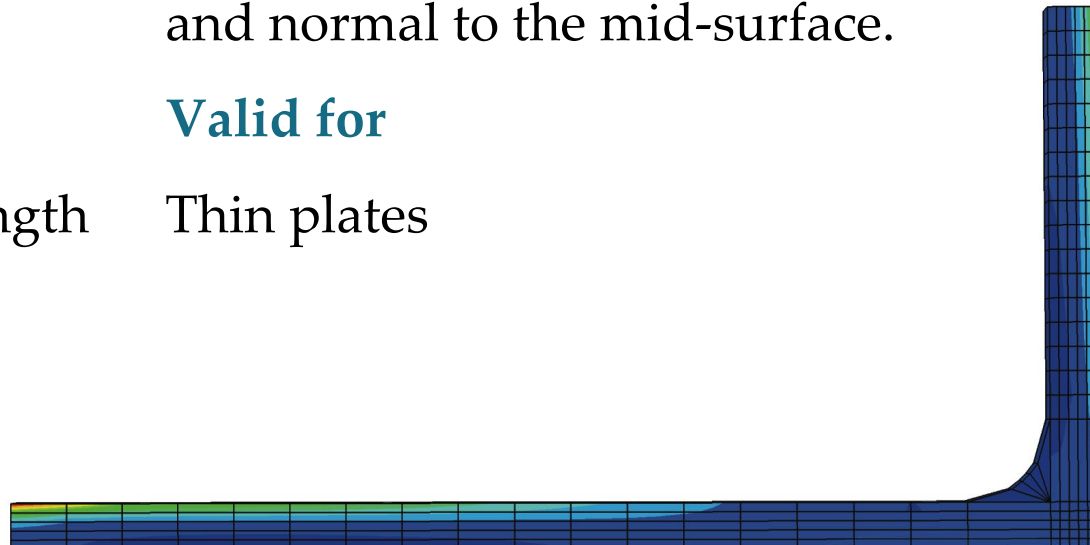


Plate theories

Two alternative theories exist for the bending of plates which are in complete analogy to beam theories:

Beam theory

Timoshenko

Kinematic assumptions

Plane sections that are normal to the axis of the beam remain plane but not necessarily normal to the axis.

Valid for

Beams with any height to length ratio

Plate theory

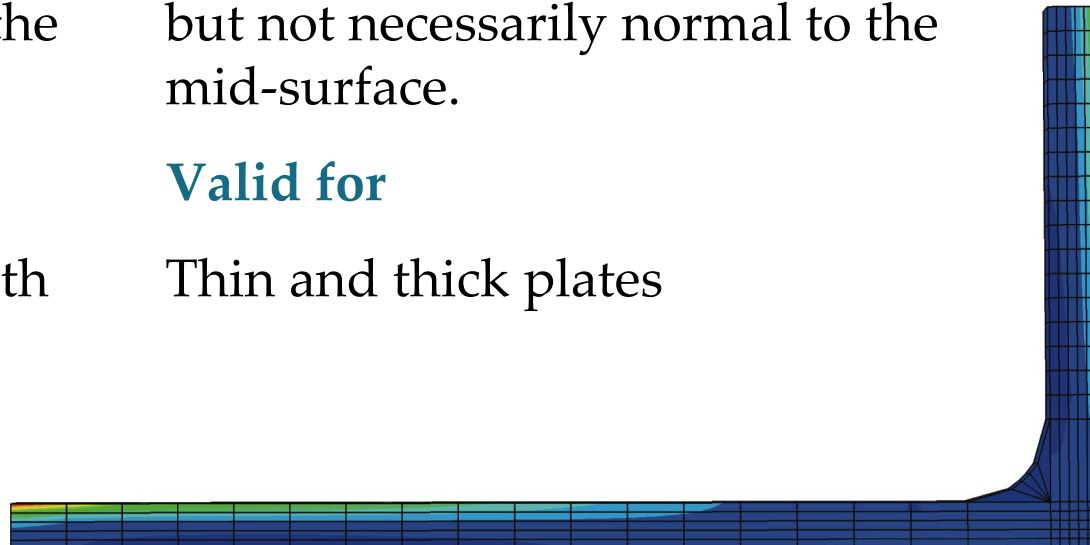
Mindlin-Reissner

Kinematic assumptions

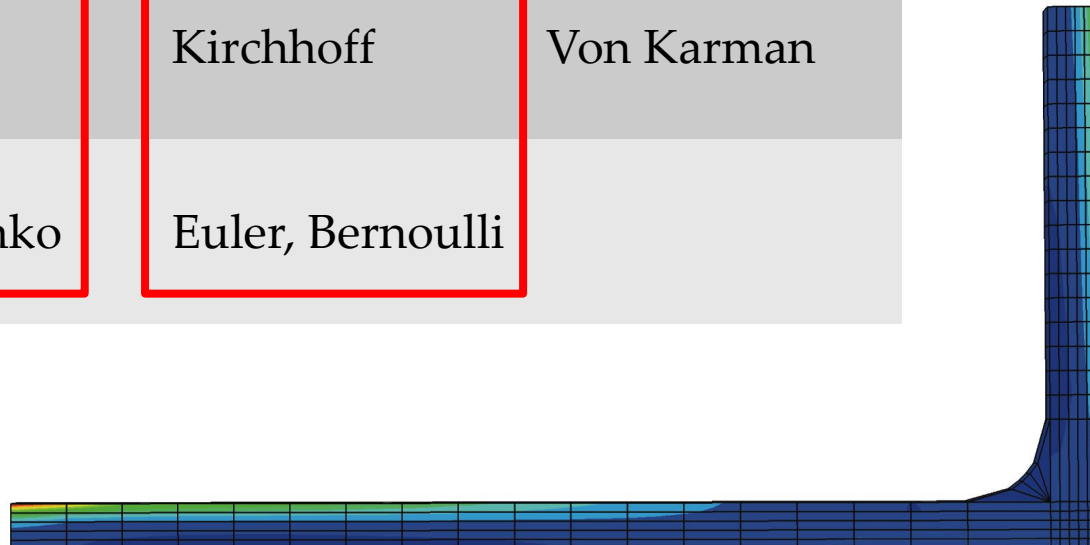
Straight lines that are normal to the mid-surface remain straight but not necessarily normal to the mid-surface.

Valid for

Thin and thick plates



	thick	thin	very thin
L/t	5-10	5-100	>100
characteristics	transverse shear deformations	no transverse shear deformations	Geometrically nonlinear
Plate theory	Reissner, Mindlin	Kirchhoff	Von Karman
Beam Theory	Timoshenko	Euler, Bernoulli	

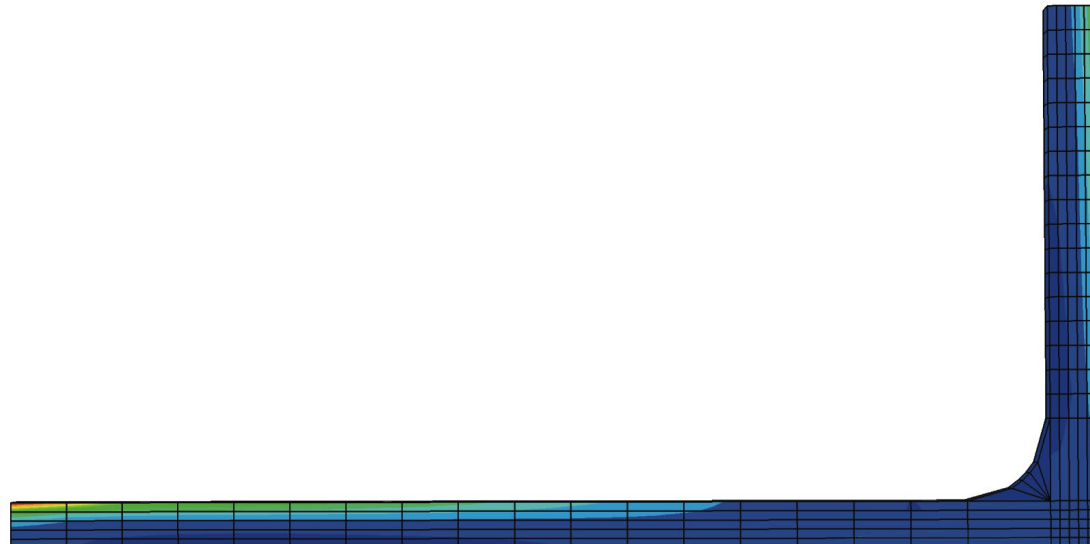


Chapter 8

The Kirchhoff-Love Plate Theory

Book Chapters

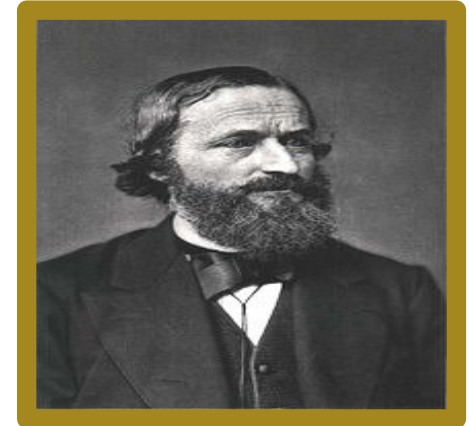
[O] V2/Ch5



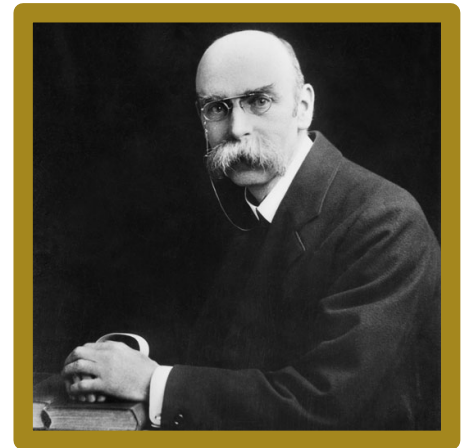
Kirchhoff-Love Plate Theory

Assumptions

- The plate is thin in the sense that the thickness t is small compared to the leading dimensions, but not so thin that the lateral deflection becomes comparable to w .
- The plate thickness is either uniform or varies slowly so that three-dimensional stress effects are ignored.
- The plate is symmetric in fabrication about the mid-surface.
- Applied transverse loads are distributed over plate surface areas of dimension t or greater.
- The support conditions are such that no significant extension of the mid-surface develops.



Gustav Robert Kirchhoff
(1824-1887)



Augustus Edward Hough Love
(1863-1940)

Kirchhoff-Love Plate Theory

Assumptions

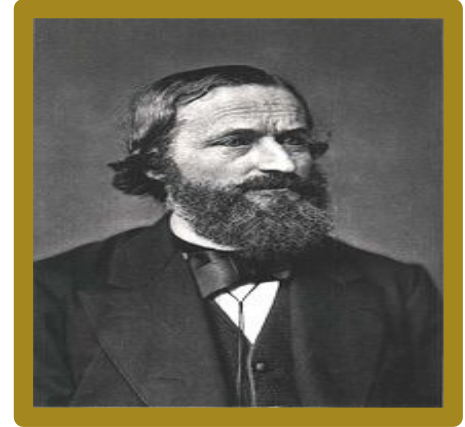
From (i), since t is very small, the variation of w with respect to z can be neglected. Therefore:

$$w(x, y, z) \approx w(x, y)$$

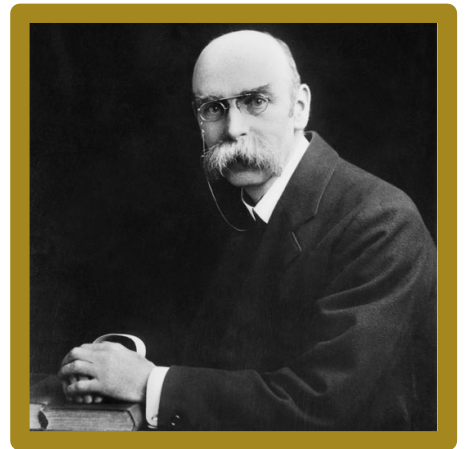
Additionally, the following kinematic assumption is introduced:

“Planes perpendicular to the mid-surface will remain plane and perpendicular to the deformed mid-surface”

This is the two-dimensional equivalent of the Euler-Bernoulli kinematic assumption for beams!!



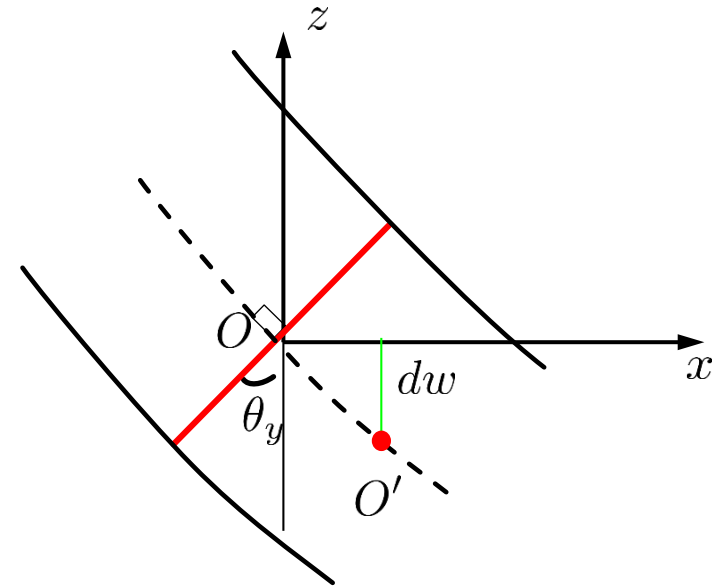
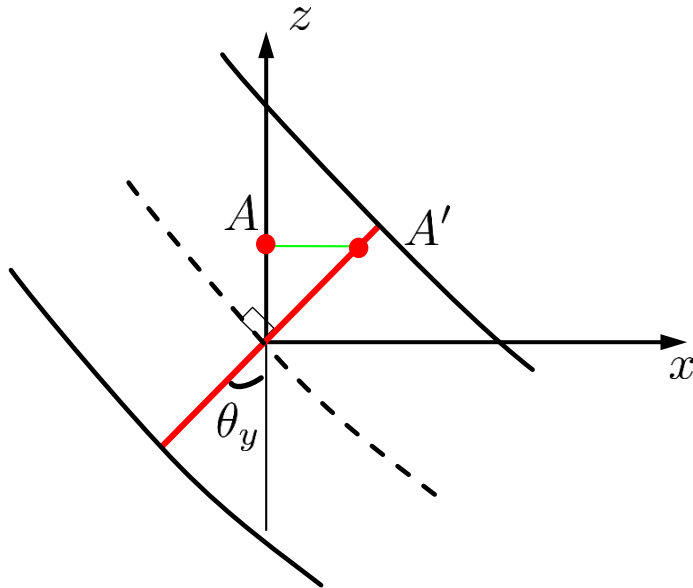
Gustav Robert Kirchhoff
(1824-1887)



Augustus Edward Hough Love
(1863-1940)

Kirchhoff-Love Plate Theory

Remember in the one-dimensional beam problem...



Point A displacement

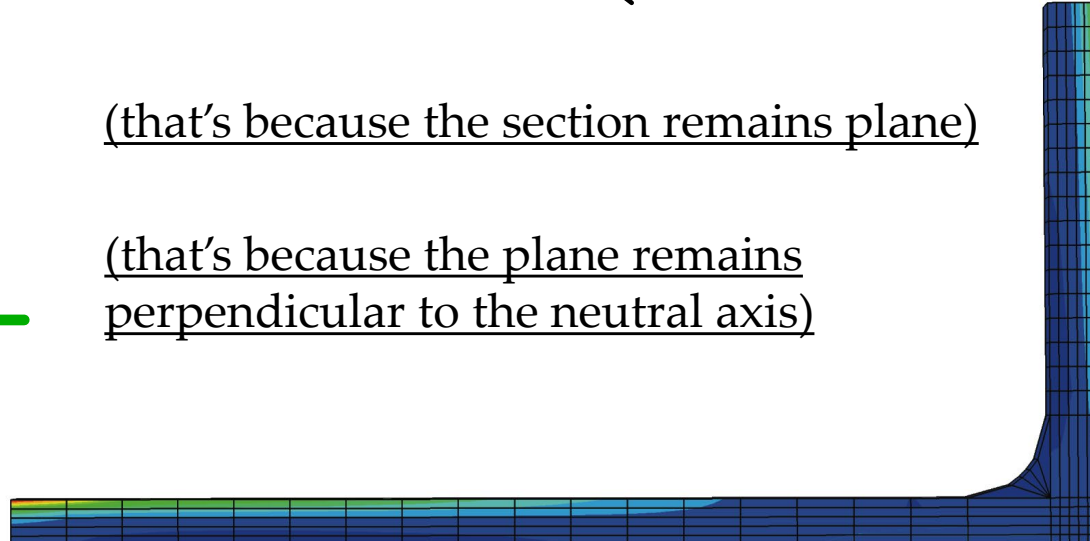
$$u_{A \rightarrow A'} = u_{bending} = z \tan(\theta_y) \approx z \theta_y$$

$$\theta_y = -\frac{dw}{dx}$$

$$u_{bending} = -z \frac{dw}{dx}$$

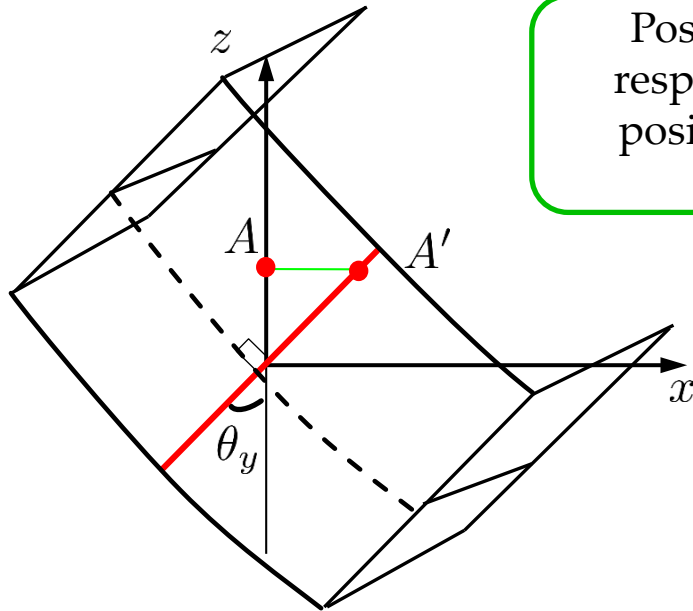
(that's because the section remains plane)

(that's because the plane remains perpendicular to the neutral axis)

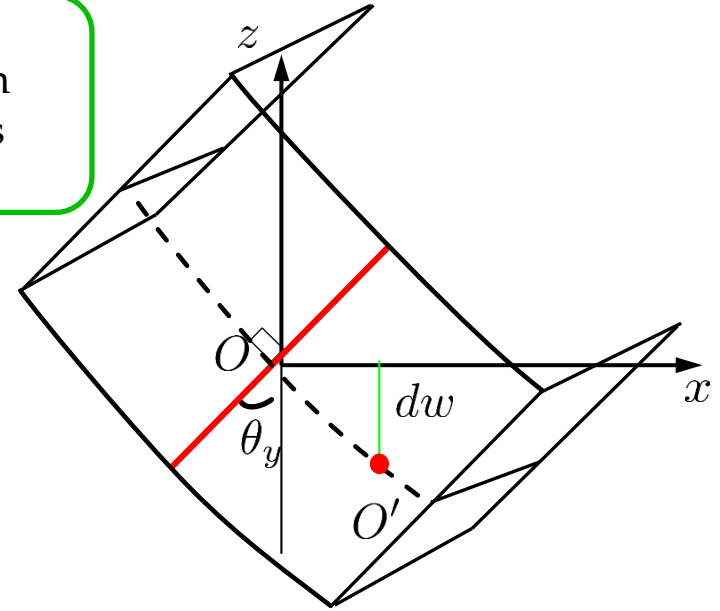


Kirchhoff-Love Plate Theory

Similarly in the two-dimensional plate problem – Bending with respect to y axis



Positive rotation with respect to y results in positive displacements along x



Point A displacement

$$u_{A \rightarrow A'} = u = z \tan(\theta_y) \approx z \theta_y$$

(that's because the section remains plane)

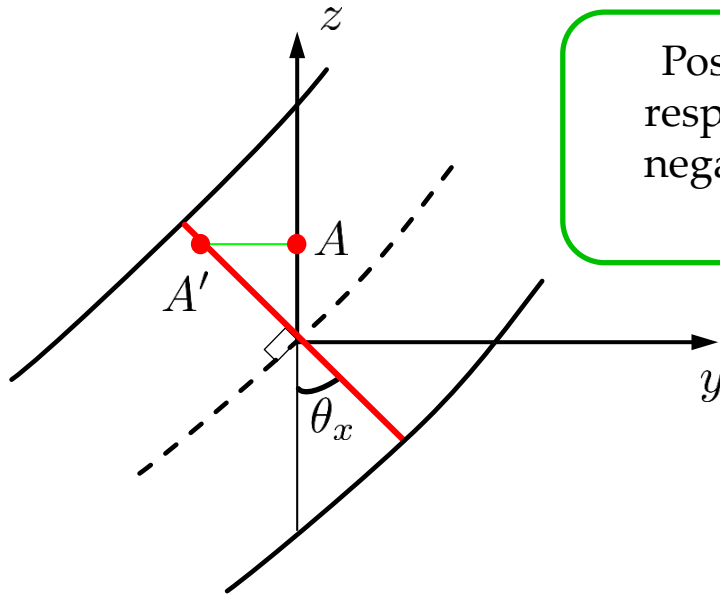
$$\theta_y = -\frac{dw}{dx}$$

(that's because the plane remains perpendicular to the neutral axis)

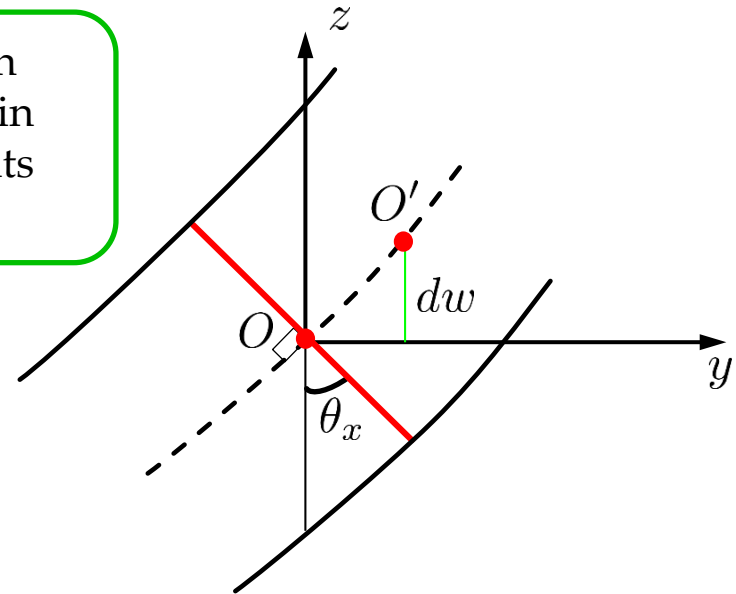
$$u = -z \frac{dw}{dx}$$

Kirchhoff-Love Plate Theory

Similarly in the two-dimensional plate problem – Bending with respect to x axis



Positive rotation with respect to x results in negative displacements along y



Point A horizontal displacement

$$v_{A \rightarrow A'} = v = -z \tan(\theta_x) \approx -z \theta_x$$

$$\theta_x = \frac{dw}{dy}$$

$$v = -z \frac{dw}{dy}$$

(that's because the section remains plane)

(that's because the plane remains perpendicular to the neutral axis)

Kirchhoff-Love Plate Theory

Compatibility Relations (Strain-Displacement)

$$\varepsilon_x = \frac{\partial u}{\partial x} = -z \frac{\partial^2 w}{\partial x^2} = -z \kappa_{xx}$$

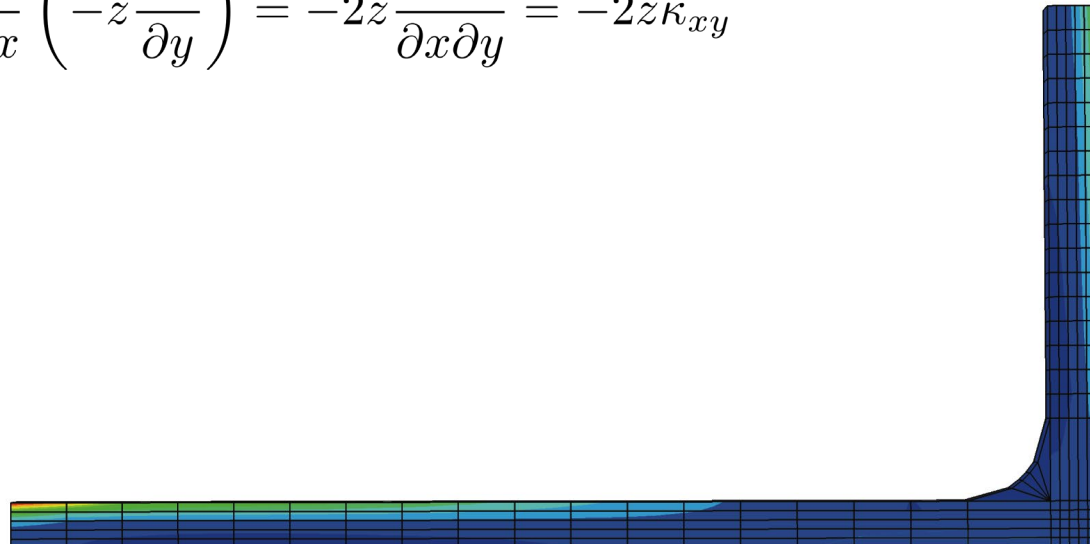
$$\varepsilon_y = \frac{\partial v}{\partial y} = -z \frac{\partial^2 w}{\partial y^2} = -z \kappa_{yy}$$

$$\varepsilon_z = \frac{\partial w}{\partial z} = 0$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \frac{\partial}{\partial y} \left(-z \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial x} \left(-z \frac{\partial w}{\partial y} \right) = -2z \frac{\partial^2 w}{\partial x \partial y} = -2z \kappa_{xy}$$

$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = -\frac{\partial w}{\partial y} + \frac{\partial w}{\partial y} = 0$$

$$\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = \frac{\partial w}{\partial x} - \frac{\partial w}{\partial x} = 0$$



Kirchhoff-Love Plate Theory

Stress-strain relations (Considering elastic isotropic material)

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} \quad \text{and} \quad \sigma_{zz} = \frac{E}{1+\nu} \left(\frac{\nu}{1-2\nu} (\varepsilon_x + \varepsilon_y) \right)$$

1st Remark

The constitutive relation results when we substitute the zero deformation terms (derived in the previous slide) to the three-dimensional elastic stress-strain relations. The derived relation is **identical** to the plane-stress case. Thus, according to the Kirchhoff-Love assumptions, every infinitesimal particle within the plate is in a plane-stress condition

2nd Remark

An immediate consequence is that the shear stress components τ_{yz} and τ_{zx} vanish

Kirchhoff-Love Plate Theory

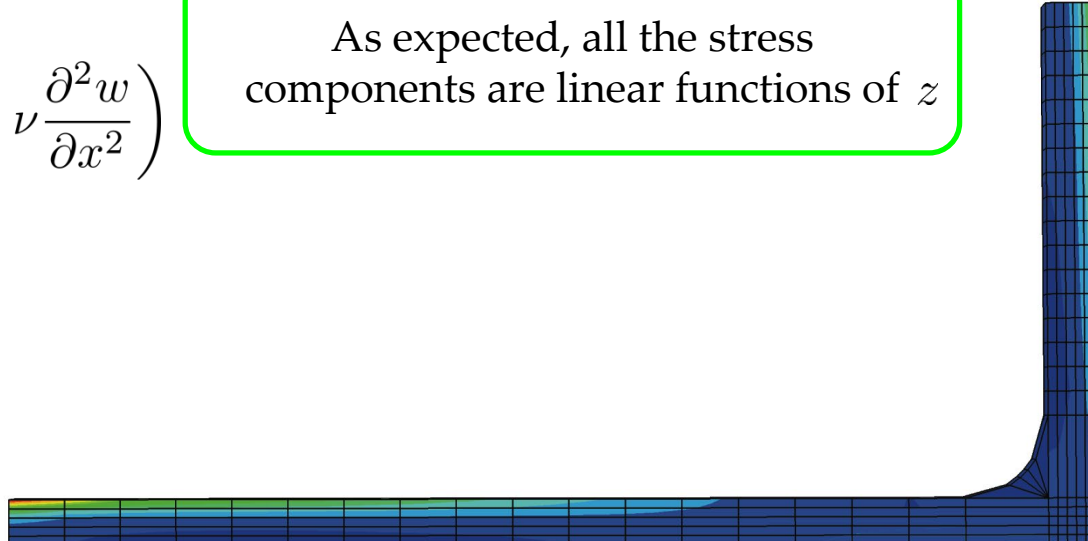
Stress-strain relations (Considering elastic isotropic material)

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} \quad \text{and} \quad \sigma_{zz} = \frac{E}{1+\nu} \left(\frac{\nu}{1-2\nu} (\varepsilon_x + \varepsilon_y) \right)$$

Substituting for the strain components with respect to the displacement

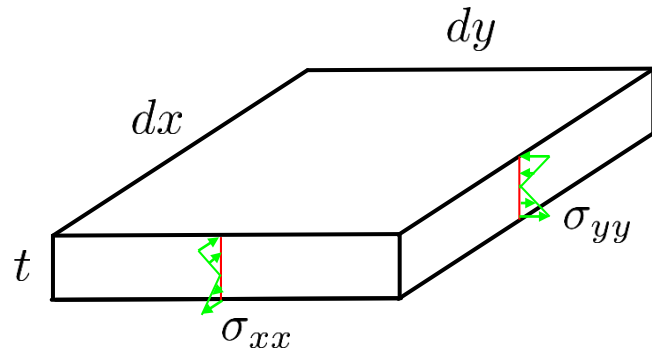
$$(I) \left\{ \begin{array}{l} \sigma_{xx} = -\frac{E}{1-\nu^2} z \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \\ \sigma_{yy} = -\frac{E}{1-\nu^2} z \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \\ \tau_{xy} = -\frac{E}{1+\nu} z \frac{\partial^2 w}{\partial x \partial y} \end{array} \right.$$

As expected, all the stress components are linear functions of z



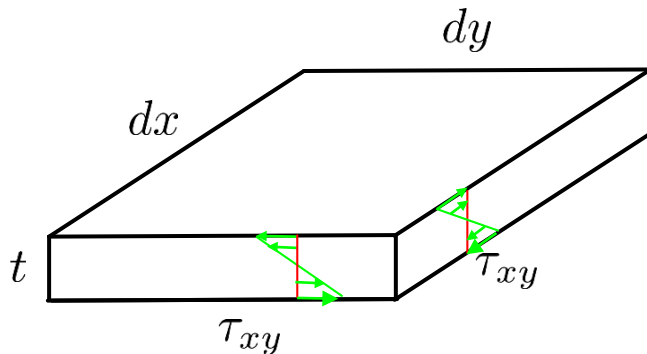
Kirchhoff-Love Plate Theory

Stress Resultants



Since the normal stress distribution is not uniform they give rise to a moment vector (created from the couple of tension and compression forces).

$$M_x = \int_{-t/2}^{t/2} -\sigma_{xx} z dz \quad M_y = \int_{-t/2}^{t/2} -\sigma_{yy} z dz$$



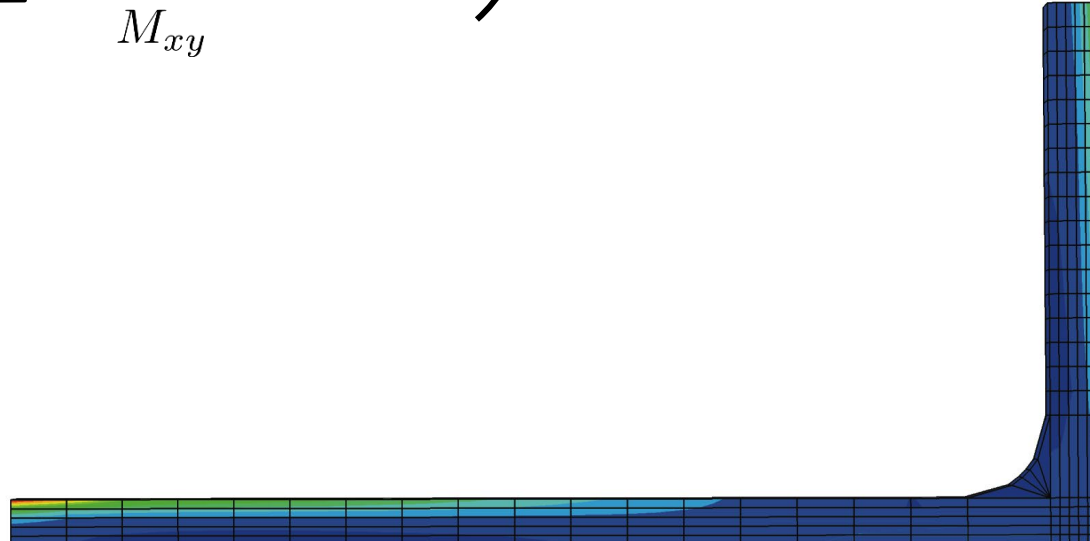
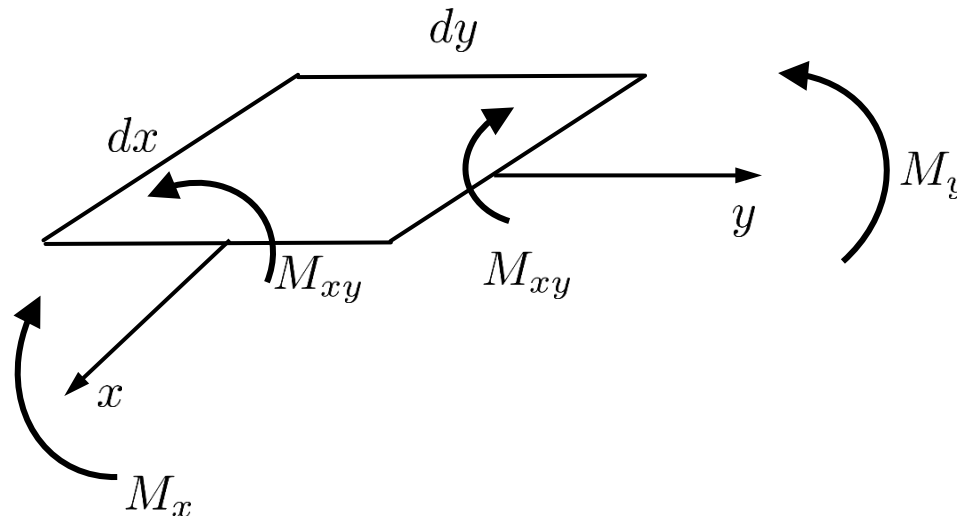
Since the shear stress distribution is not uniform they also give rise to a twisting moment vector (created from the couple of tension and compression forces).

$$M_{xy} = \int_{-t/2}^{t/2} -\sigma_{xy} z dz$$

Kirchhoff-Love Plate Theory

Stress Resultants

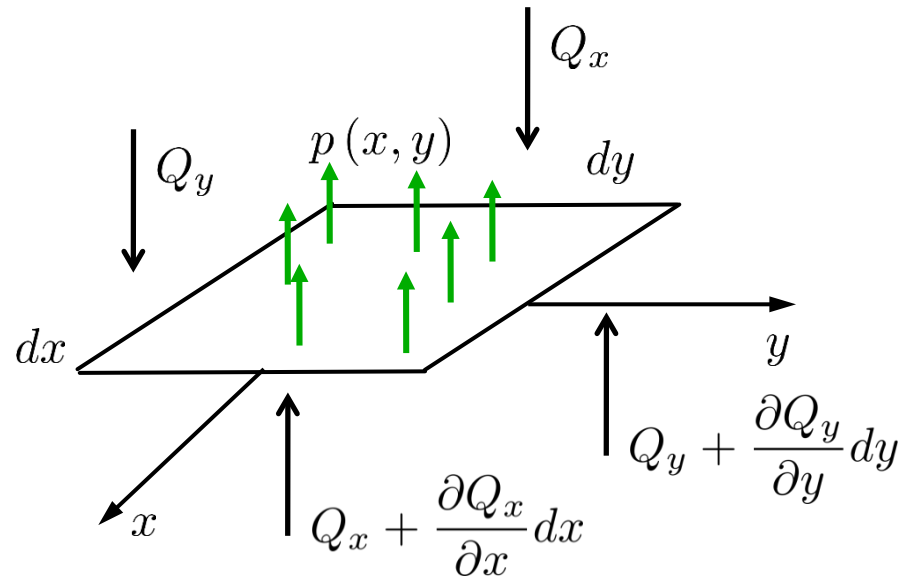
However, M_x, M_y, M_{xy} are not the only stress-resultants. For these moments to be in equilibrium, a pair of shear forces must exist. Considering an infinitesimal mid-surface element $dx dy$ the positive moments are defined as:



Kirchhoff-Love Plate Theory

Stress Resultants

If a distributed load is applied onto the element, equilibrium with respect to the vertical axis z results in:



$$\sum F_z = 0 \Rightarrow$$

$$\left(Q_y + \frac{\partial Q_y}{\partial y} dy - Q_y \right) dy + \left(Q_x + \frac{\partial Q_x}{\partial x} dx - Q_x \right) dx + p(x, y) dx dy = 0 \Rightarrow$$

$$\frac{\partial Q_y}{\partial y} + \frac{\partial Q_x}{\partial x} = -p(x, y)$$

Kirchhoff-Love Plate Theory

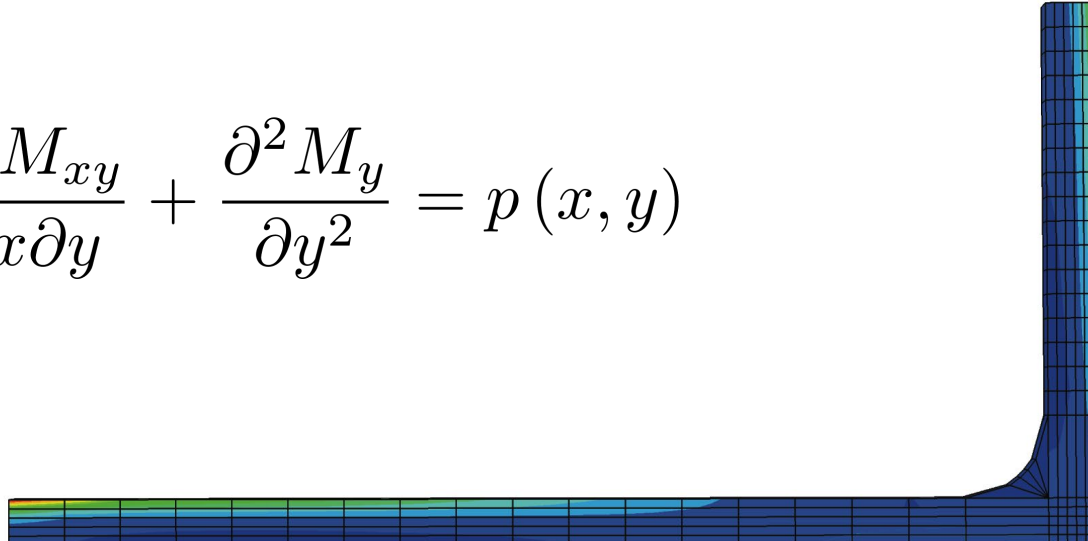
Stress Resultants

Similarly, the following equations are derived, considering moment equilibrium with respect to x, y, z axes

$$\frac{\partial M_x}{\partial x} + \frac{\partial M_{xy}}{\partial y} = -Q_x \quad \longrightarrow \quad \frac{\partial^2 M_x}{\partial x^2} + \frac{\partial^2 M_{xy}}{\partial x \partial y} = -\frac{\partial Q_x}{\partial x}$$

$$\frac{\partial M_y}{\partial y} + \frac{\partial M_{xy}}{\partial x} = -Q_y \quad \longrightarrow \quad \frac{\partial^2 M_y}{\partial y^2} + \frac{\partial^2 M_{xy}}{\partial x \partial y} = -\frac{\partial Q_y}{\partial y}$$

$$\frac{\partial^2 M_x}{\partial x^2} + 2 \frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = p(x, y)$$



Kirchhoff-Love Plate Theory

Differential Form with respect to w

Substituting the stress-displacement relations (I) into the definition of the moment components the following relations are derived

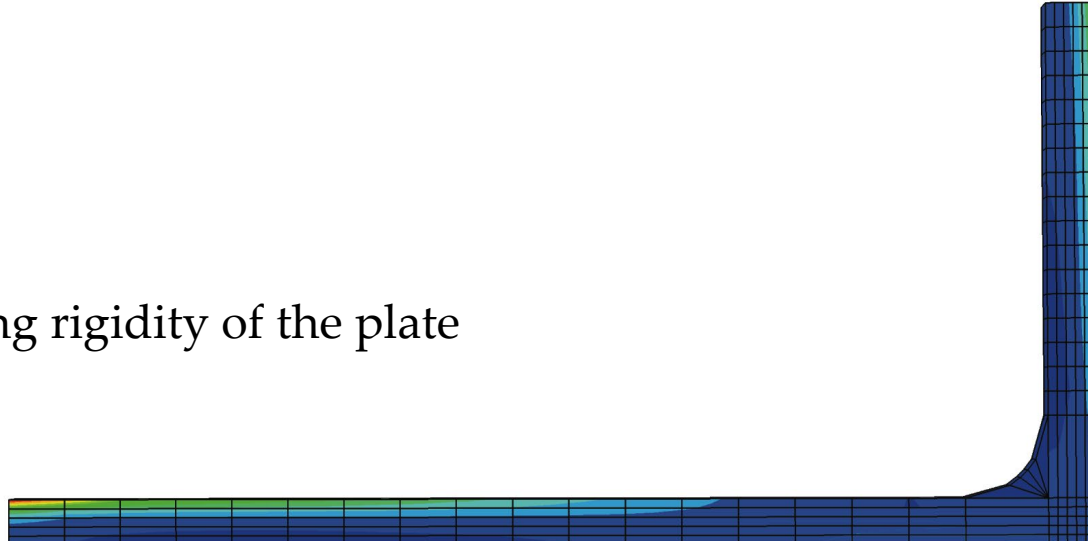
$$M_x = -D_k \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right)$$

$$M_y = -D_k \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right)$$

$$M_{xy} = -D_k (1 - \nu) \frac{\partial^2 w}{\partial x \partial y}$$

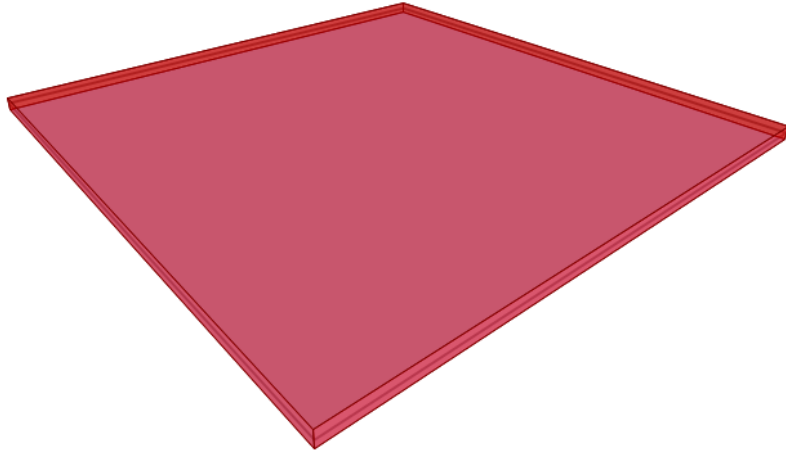
$$D_k = \frac{E}{1 - \nu^2} \int_{-t/2}^{t/2} z^2 dz \quad \text{the bending rigidity of the plate}$$

$$\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} = \frac{p(x, y)}{D_k}$$



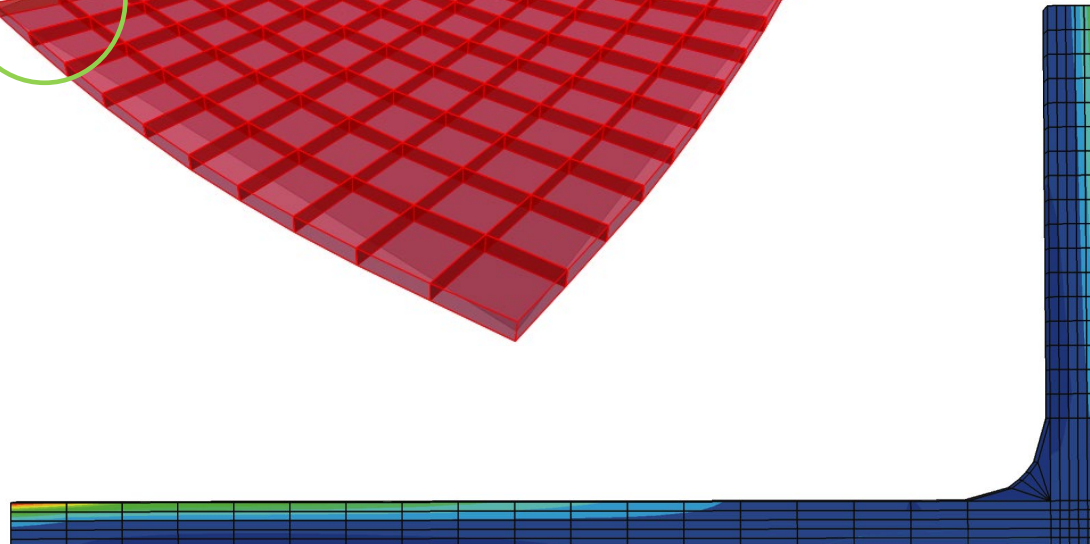
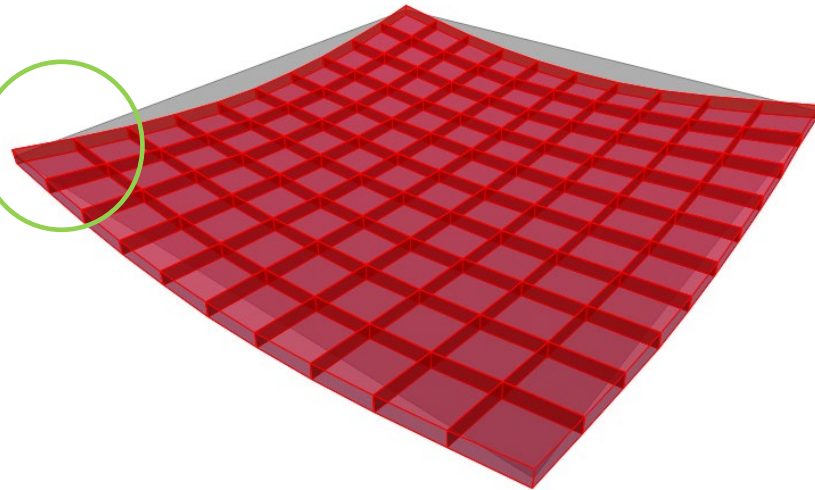
Kirchhoff-Love Plate Theory

Square simply supported plate subject to uniform distributed loading



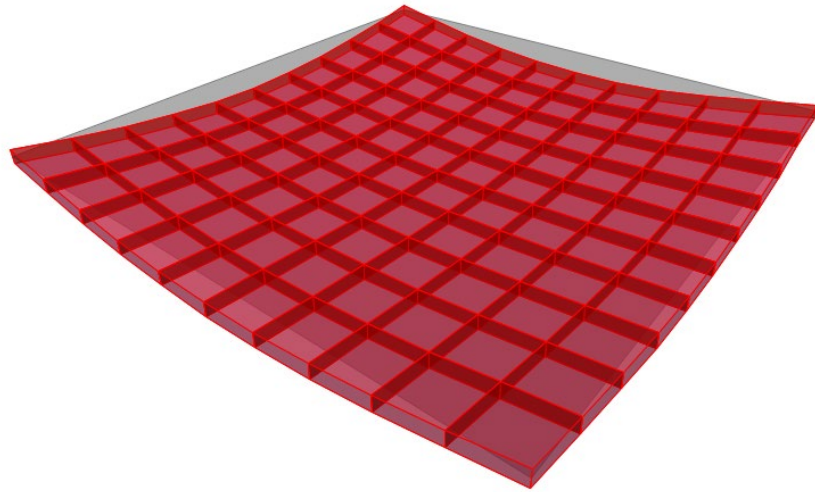
bending in both directions

pay attention to the way the corner regions deform....



Kirchhoff-Love Plate Theory

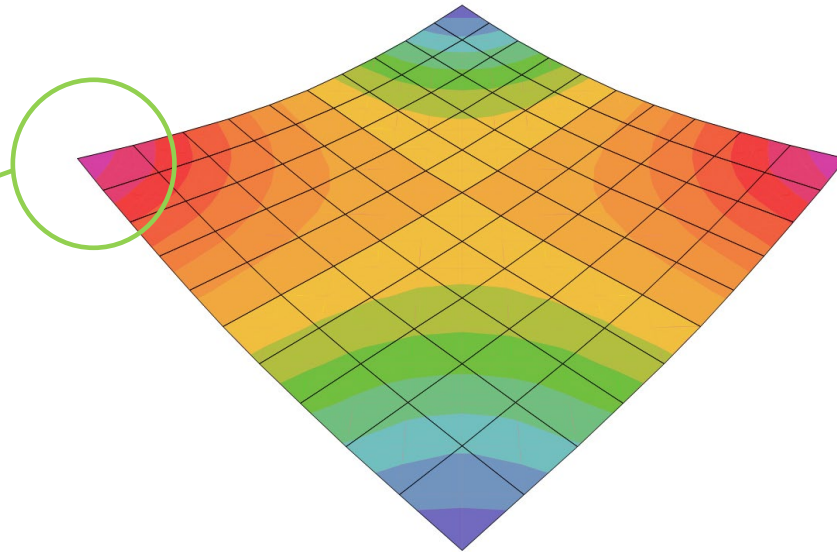
This is the effect of the twisting moment M_{xy}



and this is why specifications make sure that an additional amount of reinforcement is provided for corner areas of slabs

bending in both directions

darker areas denote larger values of twisting moments



Kirchhoff-Love Plate Theory

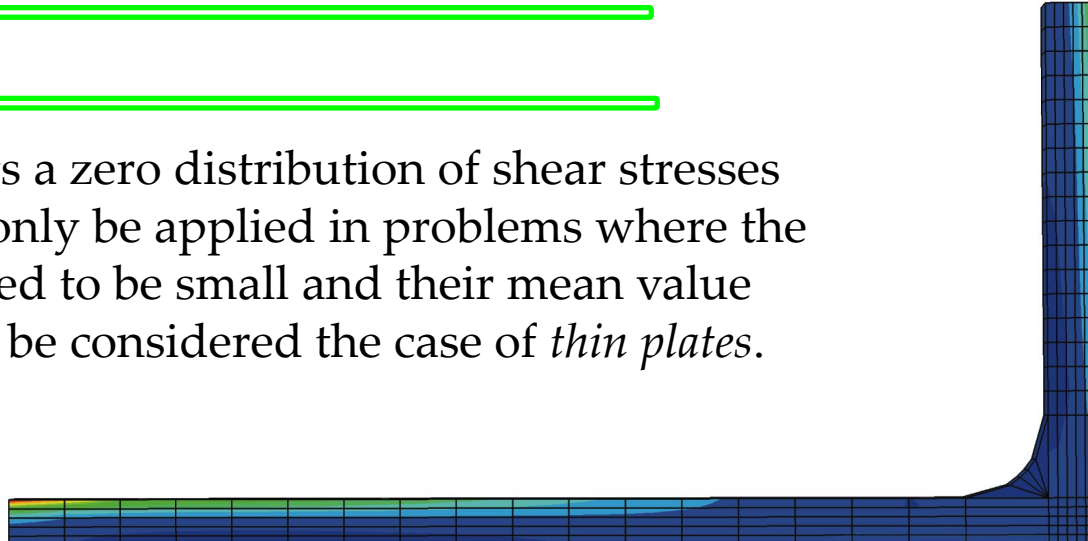
- extends the Euler/Bernoulli beam assumptions to the two-dimensional case
- Based on that, every significant measure of rotation, force, moment is evaluated with respect to the vertical deflection

Assumptions:

The main kinematic assumption is that *“Plane surfaces remain plane and perpendicular to the mid-surface of the plate”*.

General Remarks

The Kirchhoff-Love theory predicts a zero distribution of shear stresses along the z direction. Thus, it can only be applied in problems where the variation of such stresses is expected to be small and their mean value does not deviate from 0. Such can be considered the case of *thin plates*.

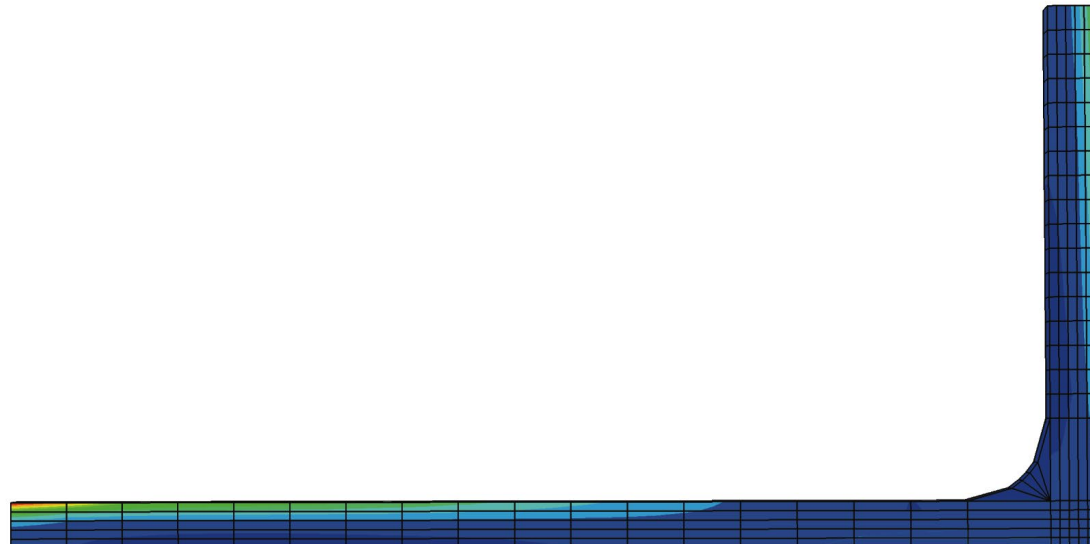


Chapter 8

The Mindlin-Reissner plate theory

Book Chapters

[O] V2/Ch5

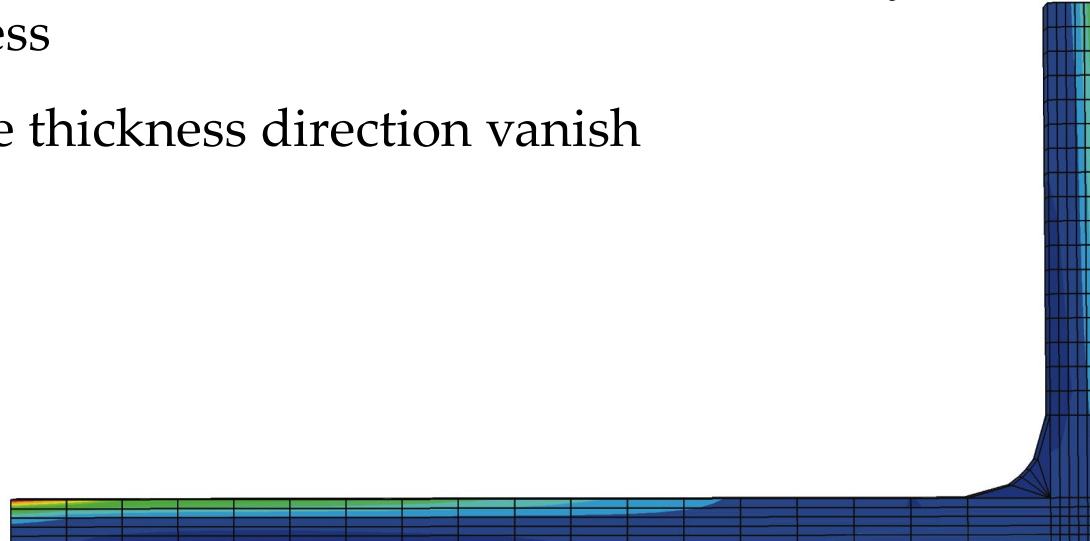


Mindlin-Reissner plate theory

Most elements used in practice are based on that theory.

The basic equations for this theory can be derived from the 3D elasticity equations by employing the following assumptions:

- Deflections don't vary along the thickness direction
- In plane displacements, and as a result strains and stresses, vary linearly along the thickness
- Normal stresses along the thickness direction vanish



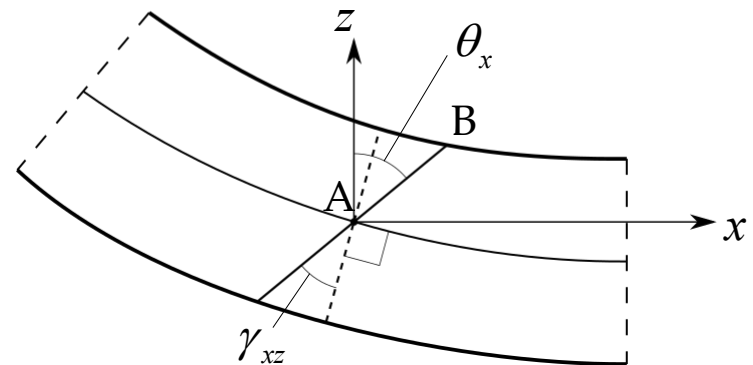
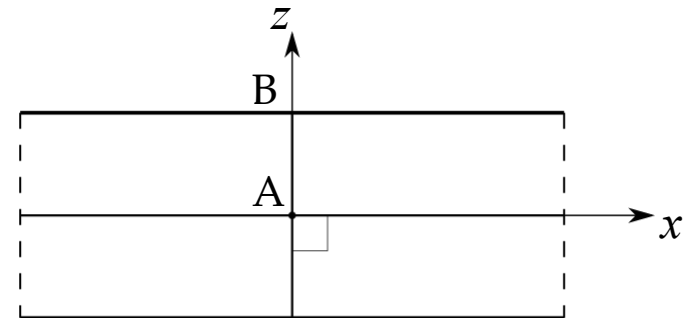
Mindlin-Reissner plate theory

The kinematic assumptions can be written as:

$$u_x = -z \theta_x(x, y); \quad u_y = -z \theta_y(x, y); \quad u_z = u_z(x, y)$$

where θ_x, θ_y are the rotations of the normal to the mid-plane along the xz and yz planes

- It should be noted that in this theory rotations are not defined as the derivatives of the deflections
- Instead they are independent variables



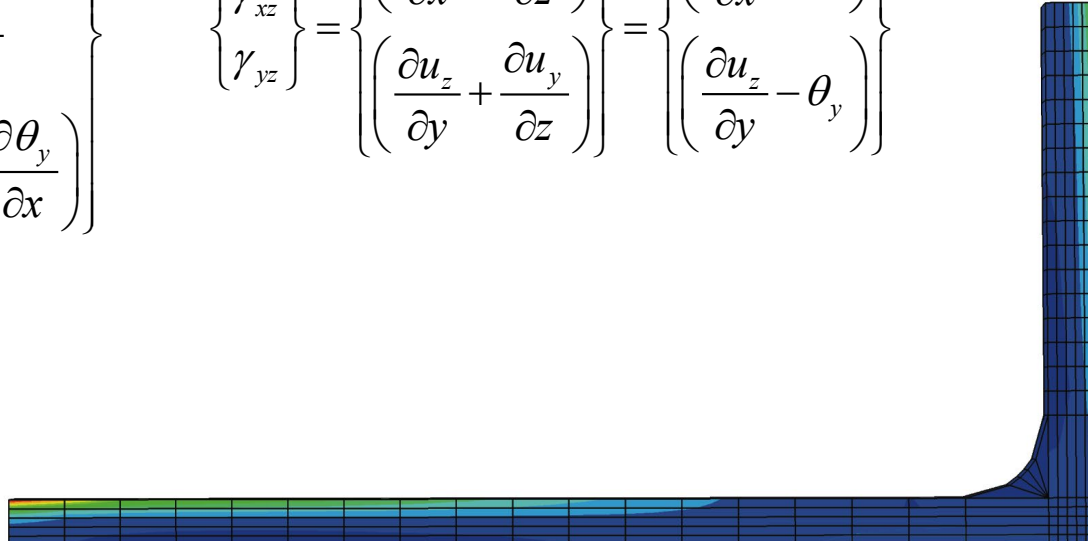
Mindlin-Reissner plate theory

The kinematic assumptions can be written as:

$$u_x = -z \theta_x(x, y); \quad u_y = -z \theta_y(x, y); \quad u_z = u_z(x, y)$$

Substituting in the strain definition and separating the in-plane from the transverse shear components:

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u_x}{\partial x} \\ \frac{\partial u_y}{\partial y} \\ \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \end{Bmatrix} = -z \begin{Bmatrix} \frac{\partial \theta_x}{\partial x} \\ \frac{\partial \theta_y}{\partial y} \\ \left(\frac{\partial \theta_x}{\partial y} + \frac{\partial \theta_y}{\partial x} \right) \end{Bmatrix} \quad \begin{Bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{Bmatrix} = \begin{Bmatrix} \left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \right) \\ \left(\frac{\partial u_z}{\partial y} + \frac{\partial u_y}{\partial z} \right) \end{Bmatrix} = \begin{Bmatrix} \left(\frac{\partial u_z}{\partial x} - \theta_x \right) \\ \left(\frac{\partial u_z}{\partial y} - \theta_y \right) \end{Bmatrix}$$

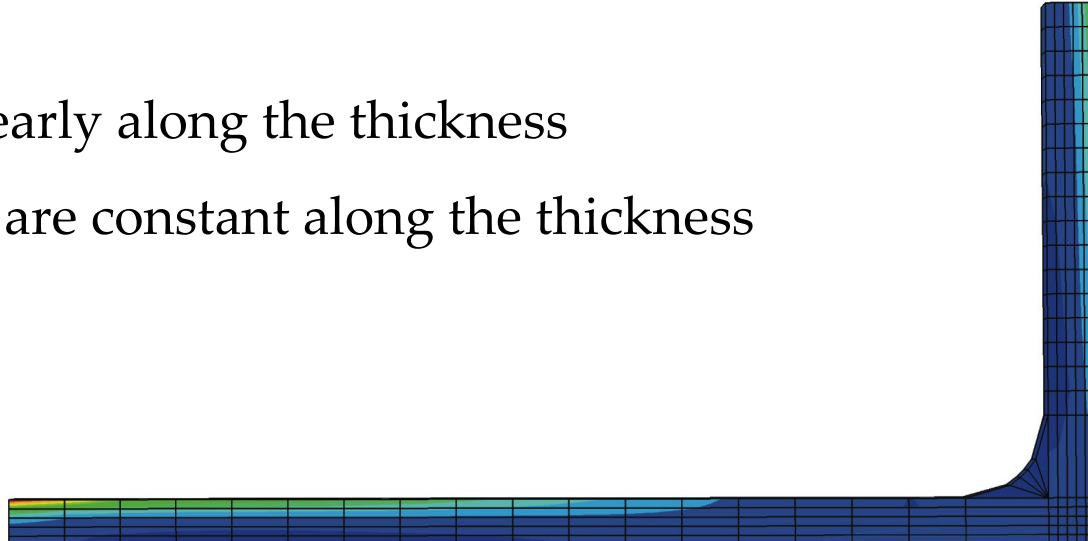


Mindlin-Reissner plate theory

Using plane stress constitutive relations, the stresses can be obtained:

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = -z \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \frac{\partial \theta_x}{\partial x} \\ \frac{\partial \theta_y}{\partial y} \\ \left(\frac{\partial \theta_x}{\partial y} + \frac{\partial \theta_y}{\partial x} \right) \end{Bmatrix} \quad \begin{Bmatrix} \tau_{xz} \\ \tau_{yz} \end{Bmatrix} = \frac{E}{2(1+\nu)} \begin{Bmatrix} \left(\frac{\partial u_z}{\partial x} - \theta_x \right) \\ \left(\frac{\partial u_z}{\partial y} - \theta_y \right) \end{Bmatrix}$$

- ✓ In plane stresses vary linearly along the thickness
- ✓ Transverse shear stresses are constant along the thickness



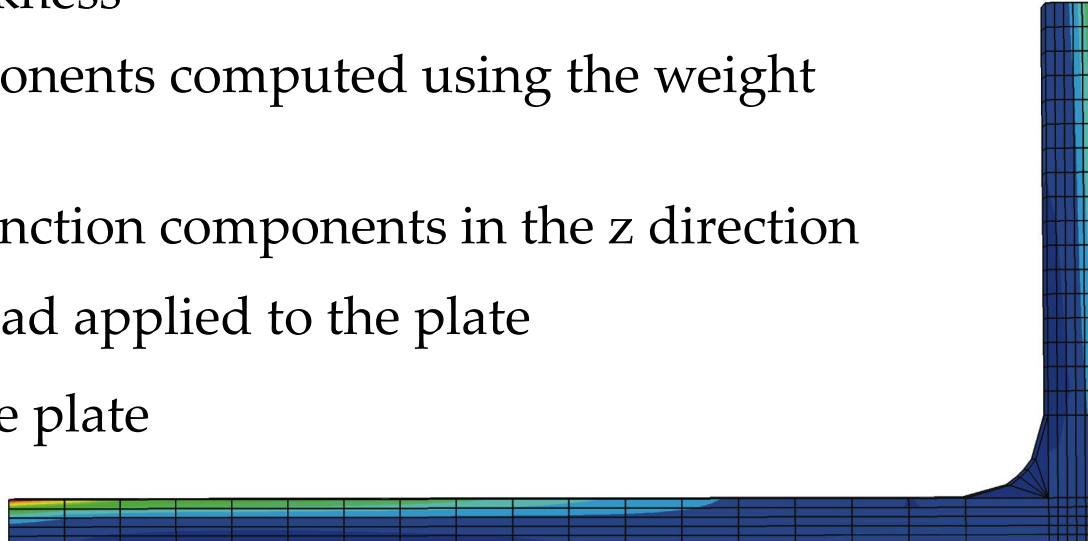
Mindlin-Reissner plate theory

By separating integration along the thickness and on the area, the weak form can be written as:

$$\frac{E}{1-\nu^2} \int_{-t/2}^{t/2} \int_A \begin{bmatrix} \bar{\varepsilon}_x & \bar{\varepsilon}_y & \bar{\gamma}_{xy} \end{bmatrix} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} dA dz + \frac{E}{2(1+\nu)} \int_{-t/2}^{t/2} \int_A \begin{bmatrix} \bar{\gamma}_{xz} & \bar{\gamma}_{yz} \end{bmatrix} \begin{Bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{Bmatrix} dA dz = \int_A w_z p_z dA$$

where:

- t is the plate thickness
- $\bar{\varepsilon}_x, \bar{\varepsilon}_y, \bar{\gamma}_{xy}, \bar{\gamma}_{xz}, \bar{\gamma}_{yz}$ are strain components computed using the weight function
- w_z is the weight function components in the z direction
- p_z is the normal load applied to the plate
- A is the area of the plate



Mindlin-Reissner plate theory

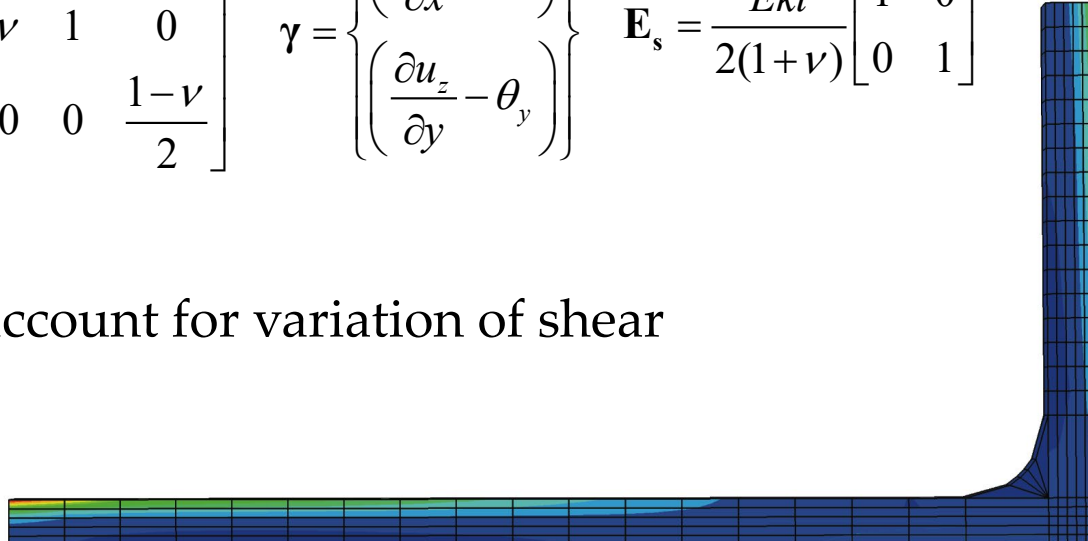
Once integration along the thickness has been carried out, the weak form can be re-written as:

$$\int_A {}^{-T}\mathbf{\kappa} \mathbf{E}_b \mathbf{\kappa} dA + \int_A {}^{-T}\boldsymbol{\gamma} \mathbf{E}_s \boldsymbol{\gamma} dA = \int_A w_z p_z dA$$

with:

$$\mathbf{\kappa} = \left\{ \begin{array}{c} \frac{\partial \theta_x}{\partial x} \\ \frac{\partial \theta_y}{\partial y} \\ \left(\frac{\partial \theta_x}{\partial y} + \frac{\partial \theta_y}{\partial x} \right) \end{array} \right\} \quad \mathbf{E}_b = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \quad \boldsymbol{\gamma} = \left\{ \begin{array}{c} \left(\frac{\partial u_z}{\partial x} - \theta_x \right) \\ \left(\frac{\partial u_z}{\partial y} - \theta_y \right) \end{array} \right\} \quad \mathbf{E}_s = \frac{Ekt}{2(1+\nu)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

k is a corrective factor to account for variation of shear stresses, usually $k = 5/6$

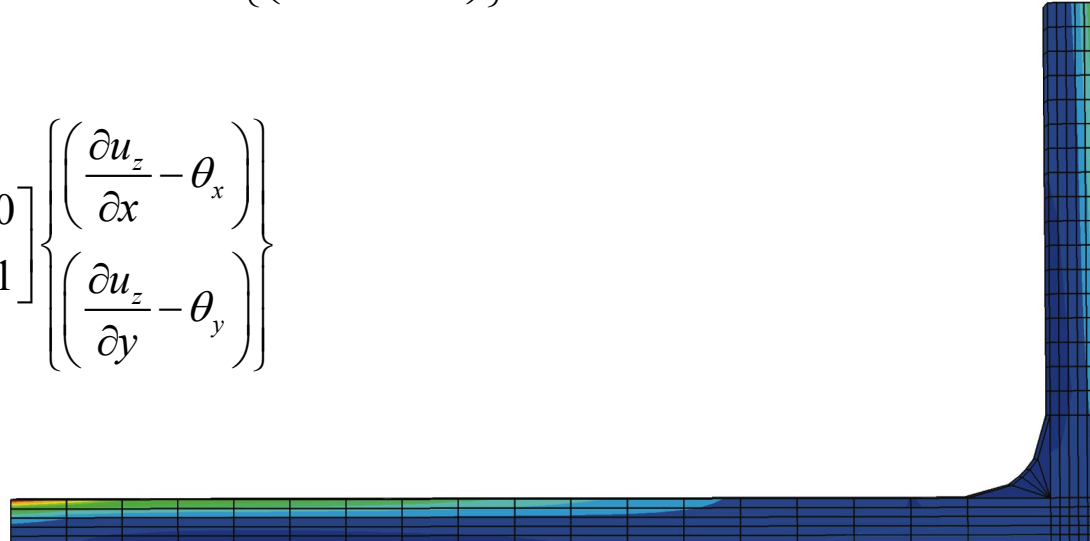


Mindlin-Reissner plate theory

Moments and shear forces are obtained from the curvatures and transverse shear stresses through the constitutive equations:

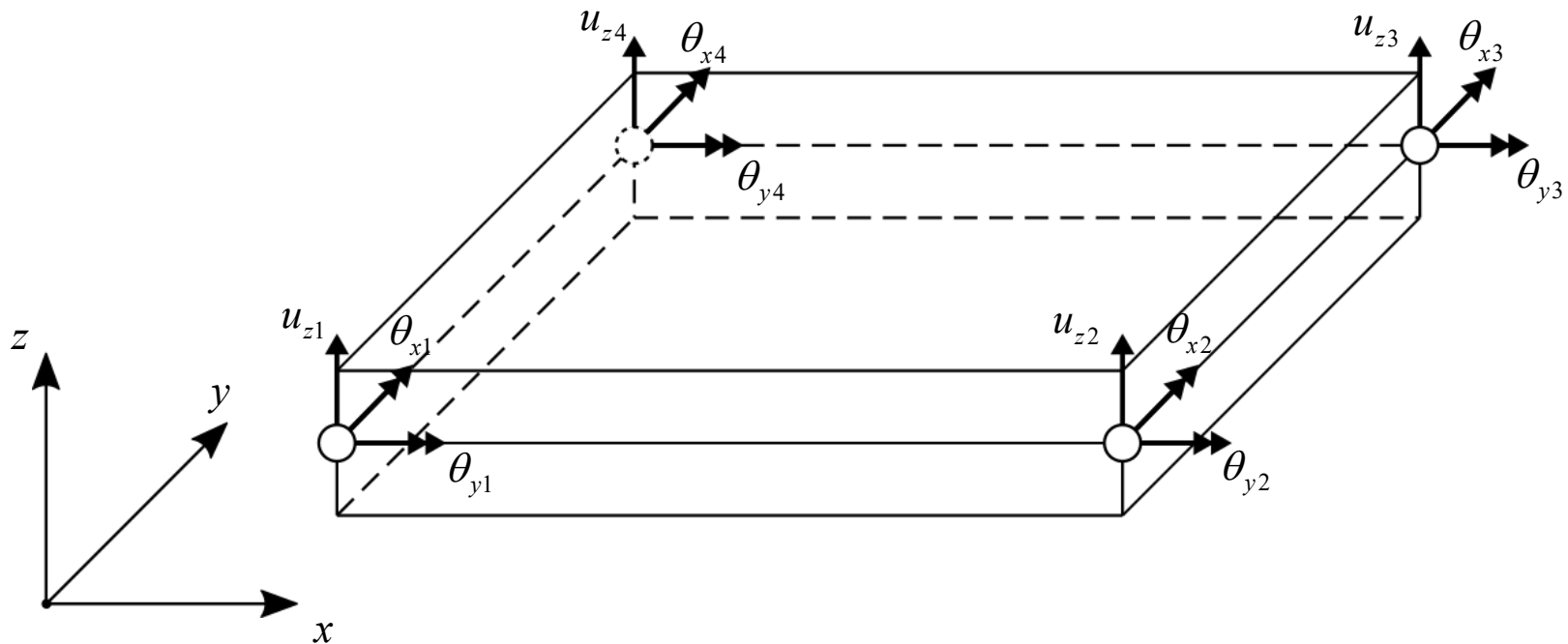
$$\mathbf{M} = \mathbf{E}_b \boldsymbol{\kappa}, \quad \begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \frac{Et^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{Bmatrix} \frac{\partial \theta_x}{\partial x} \\ \frac{\partial \theta_y}{\partial y} \\ \left(\frac{\partial \theta_x}{\partial y} + \frac{\partial \theta_y}{\partial x} \right) \end{Bmatrix}$$

$$\mathbf{Q} = \mathbf{E}_s \boldsymbol{\gamma}, \quad \begin{Bmatrix} Q_x \\ Q_y \end{Bmatrix} = \frac{Et}{2(1+\nu)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \left(\frac{\partial u_z}{\partial x} - \theta_x \right) \\ \left(\frac{\partial u_z}{\partial y} - \theta_y \right) \end{Bmatrix}$$



Isoparametric plate elements

Element geometry, and degree of freedom definitions:

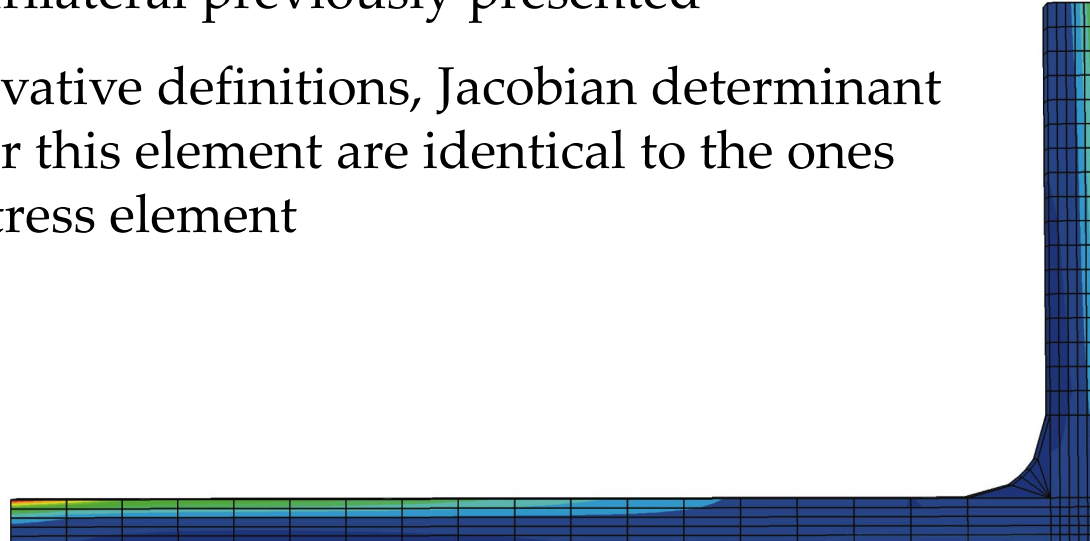


Isoparametric plate elements

Since rotations and deflections are independent, they are also discretized independently:

$$u_z = \sum_{i=1}^n h_i(r,s) u_{zi}; \quad \theta_x = \sum_{i=1}^n h_i(r,s) \theta_{xi}; \quad \theta_y = \sum_{i=1}^n h_i(r,s) \theta_{yi}$$

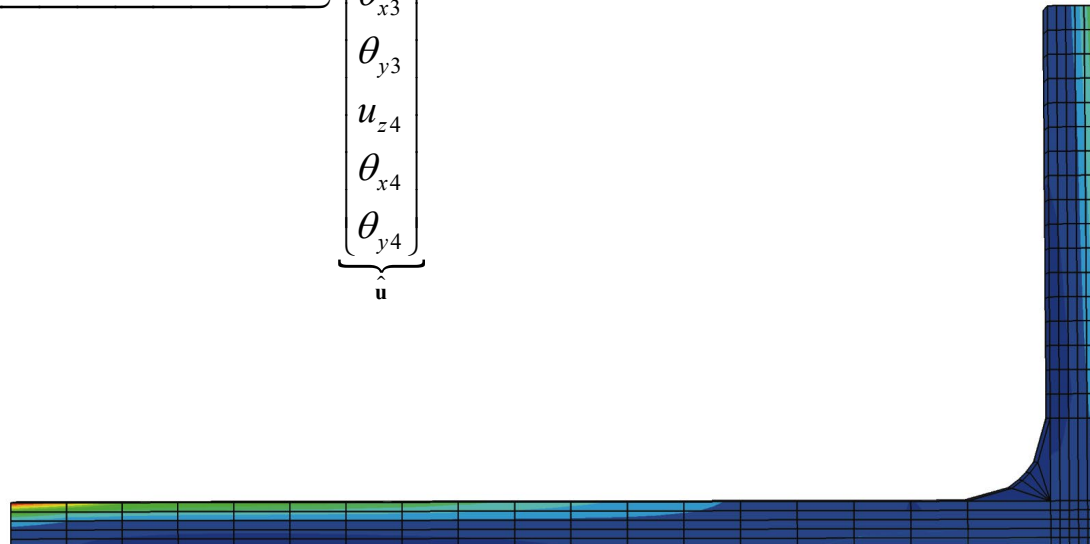
- One of the most commonly used elements for this kind of analysis is the isoparametric quadrilateral previously presented
- The shape functions, derivative definitions, Jacobian determinant and Gauss points used for this element are identical to the ones presented for the plane stress element



Isoparametric plate elements

Curvatures and shear strains can then be obtained as:

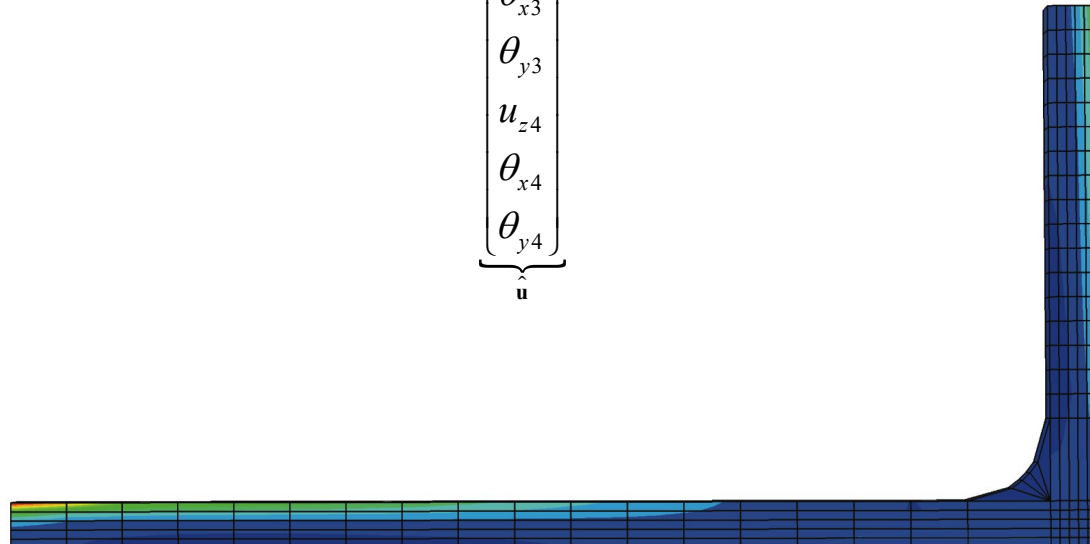
$$\mathbf{\kappa} = \underbrace{\begin{bmatrix} 0 & h_{1,x} & 0 & 0 & h_{2,x} & 0 & 0 & h_{3,x} & 0 & 0 & h_{4,x} & 0 \\ 0 & 0 & h_{1,y} & 0 & 0 & h_{1,y} & 0 & 0 & h_{1,y} & 0 & 0 & h_{1,y} \\ 0 & h_{1,y} & h_{1,x} & 0 & h_{2,y} & h_{2,x} & 0 & h_{3,y} & h_{3,x} & 0 & h_{4,y} & h_{4,x} \end{bmatrix}}_{\mathbf{B}_b} \underbrace{\begin{Bmatrix} u_{z1} \\ \theta_{x1} \\ \theta_{y1} \\ u_{z2} \\ \theta_{x2} \\ \theta_{y2} \\ u_{z3} \\ \theta_{x3} \\ \theta_{y3} \\ u_{z4} \\ \theta_{x4} \\ \theta_{y4} \end{Bmatrix}}_{\hat{\mathbf{u}}} \longrightarrow \mathbf{\kappa} = \mathbf{B}_b \hat{\mathbf{u}}$$



Isoparametric plate elements

Curvatures and shear strains can then be obtained as:

$$\gamma = \underbrace{\begin{bmatrix} h_{1,x} & -h_1 & 0 & h_{2,x} & -h_2 & 0 & h_{3,x} & -h_3 & 0 & h_{4,x} & -h_4 & 0 \\ h_{1,y} & 0 & -h_1 & h_{2,y} & 0 & -h_2 & h_{3,y} & 0 & -h_3 & h_{4,y} & 0 & -h_4 \end{bmatrix}}_{\mathbf{B}_s} \underbrace{\begin{Bmatrix} u_{z1} \\ \theta_{x1} \\ \theta_{y1} \\ u_{z2} \\ \theta_{x2} \\ \theta_{y2} \\ u_{z3} \\ \theta_{x3} \\ \theta_{y3} \\ u_{z4} \\ \theta_{x4} \\ \theta_{y4} \end{Bmatrix}}_{\hat{\mathbf{u}}} \rightarrow \gamma = \mathbf{B}_s \hat{\mathbf{u}}$$



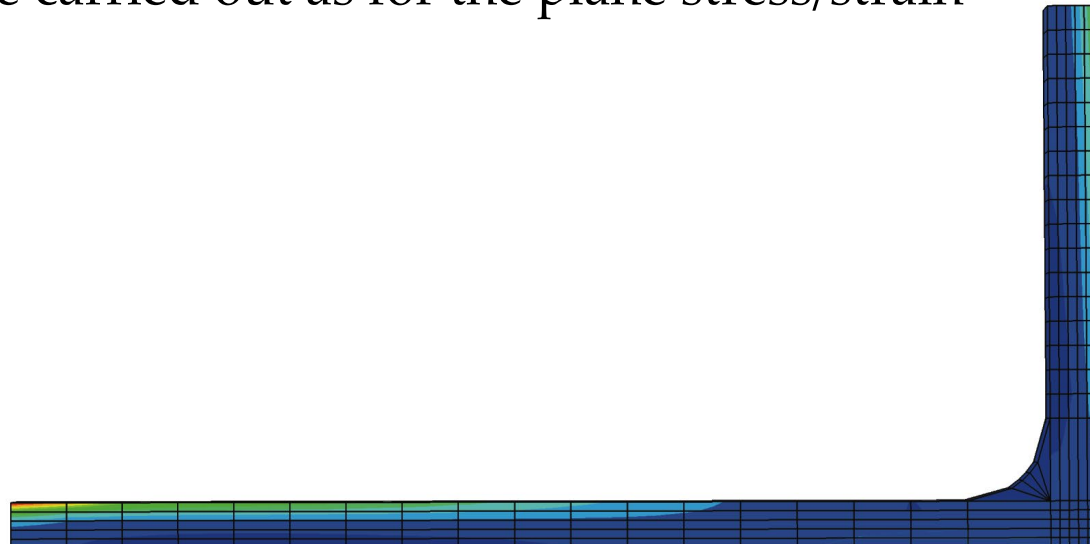
Isoparametric plate elements

The element stiffness matrix can be obtained as:

$$\mathbf{K} = \int_{-1}^1 \int_{-1}^1 \mathbf{B}_b^T \mathbf{E}_b \mathbf{B}_b \det(\mathbf{J}) dr ds + \int_{-1}^1 \int_{-1}^1 \mathbf{B}_s^T \mathbf{E}_s \mathbf{B}_s \det(\mathbf{J}) dr ds$$

$$\mathbf{f} = \int_{-1}^1 \int_{-1}^1 \mathbf{N}^T p_z \det(\mathbf{J}) dr ds$$

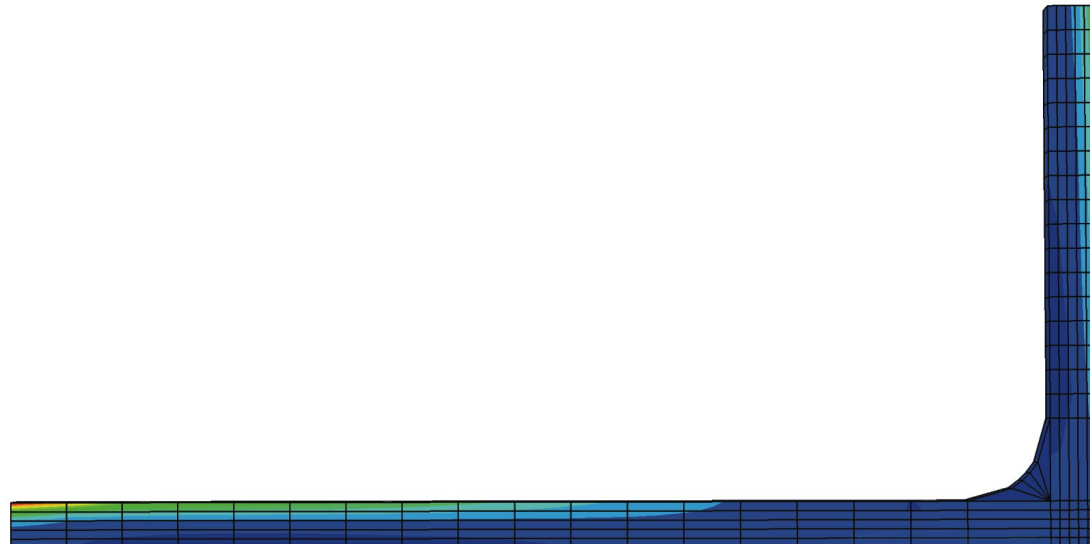
Numerical integration can be carried out as for the plane stress/strain elements.



Isoparametric plate elements

The element presented:

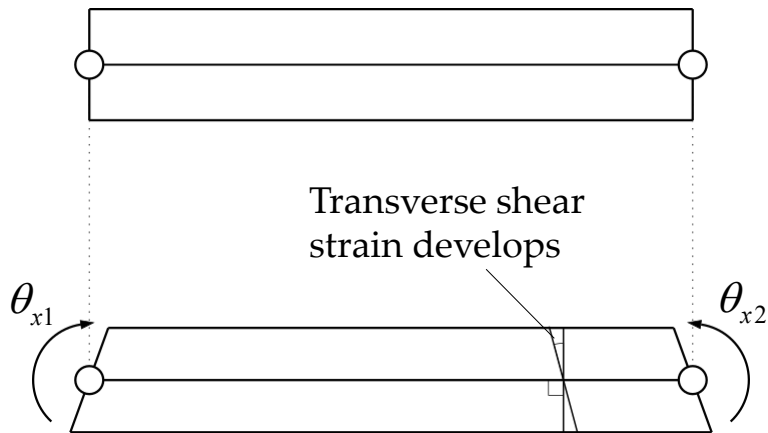
- Is simple to formulate and implement
- Offers good performance for relatively large thickness to length ratios ($>1/10$)
- Suffers from “locking” problems for smaller ratios



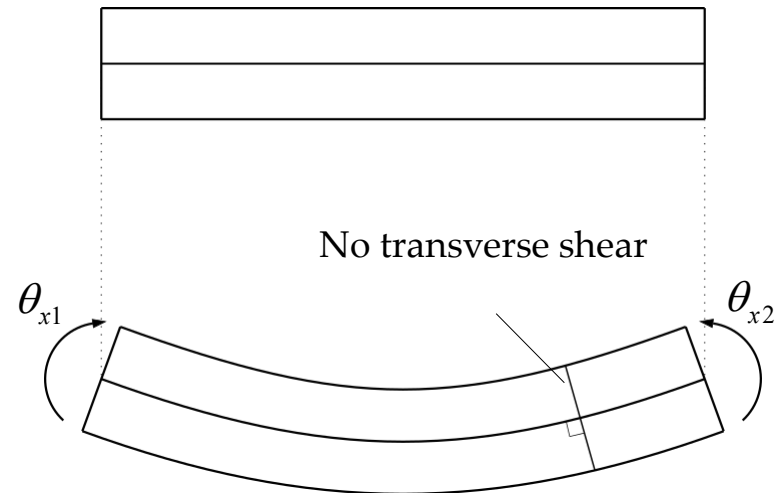
The shear locking problem

Let's consider the presented element in a state of pure bending:

Element deformation



Actual deformation
(for a thin plate)



As the thickness decreases, the “spurious” energy produced by this shear deformation dominates the total energy leading to unrealistically small deflections!

The shear locking problem

An alternative remedy: Assumed shear strain field

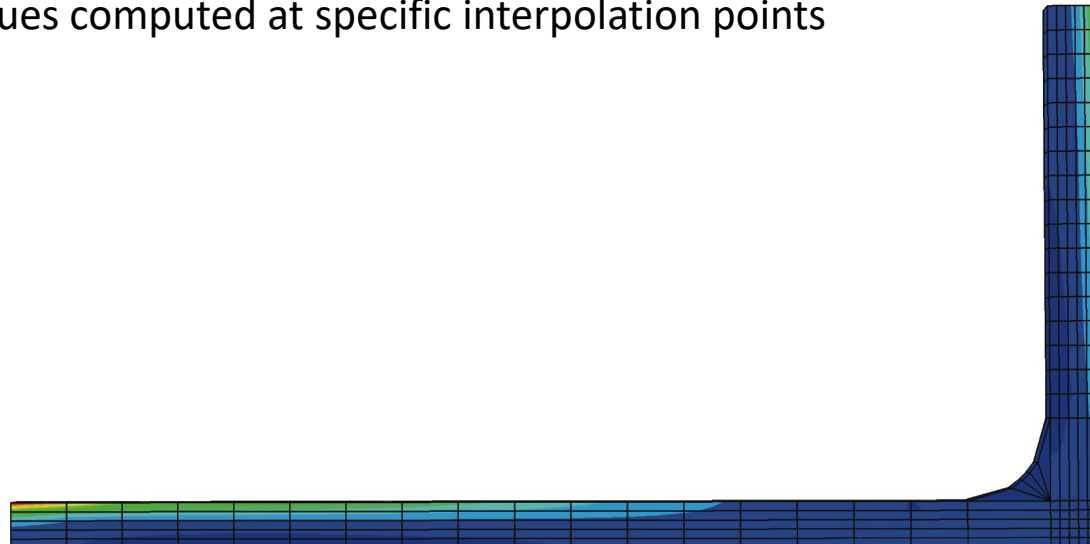
An alternative interpolation scheme can be used for the shear strain:

$$\gamma = \mathbf{N}_\gamma \gamma^{DI}$$

where:

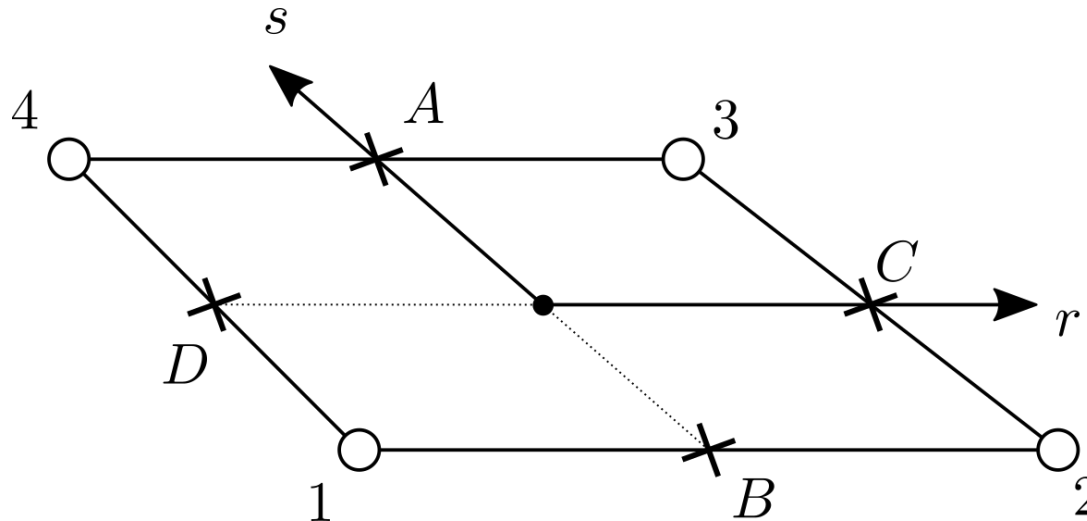
$\mathbf{N}_\gamma$ → a vector of interpolation functions used for the shear strain

γ^{DI} → a vector of shear strain values computed at specific interpolation points



The shear locking problem

Strain interpolation for linear quadrilateral element



$$e_{rt} = \frac{1}{2}(1+s)e_{rt}^{(A)} + \frac{1}{2}(1-s)e_{rt}^{(B)}$$

$$e_{st} = \frac{1}{2}(1+r)e_{st}^{(C)} + \frac{1}{2}(1-r)e_{st}^{(D)}$$

→ strain components are defined in isoparametric coordinates

→ transformation to Cartesian coordinates is required